Navigating Trends: A Comparative Study of LSTM, ARIMA, and CNN Models with Technical, Historical, and Combined Indicators in Stock Prediction

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Abstract

This thesis tackles the challenge of accurately forecasting short-term price trends in financial markets. Grounded in prior research, the central research question revolves around improving the performance of forecasting models, specifically ARIMA, LSTM, and 1D-CNN, by considering a comprehensive set of indicators, including historical and technical factors, while using different input and output lengths and evaluating performance on different sectors. This approach distinguishes itself by incorporating both historical and technical indicators, a novel feature absent in prior studies. Additionally, the unique use of varying input and output lengths and sector-specific evaluations sets this method apart, introducing unprecedented dimensions to short-term price trend forecasting. This thesis utilizes a dataset sourced directly from financial websites, allowing for the download and reuse of data. The primary findings highlight the consistent outperformance of the LSTM model, leveraging only technical indicators, compared to ARIMA and 1D-CNN models. Notably, the study reveals that increasing the output length enhances accuracy for all models, potentially allowing for a more comprehensive grasp of technical patterns. The best-performing model is identified as LSTM with technical indicators and an demonstrating its ability to generalize across different sectors within the NASDAQ100. This research provides nuanced insights into the intricate balance between model simplicity and feature relevance, offering practical guidance for robust financial trend forecasting. The research significantly advances financial market forecasting by demonstrating the efficacy of LSTM models with technical indicators. This innovation has broad implications, offering valuable insights for traders, investors, and financial analysts, enhancing decision-making processes and potentially improving overall market efficiency.

Keywords: Stock trend prediction, LSTM, ARIMA, CNN

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Data Source/Code/Ethics/Technology Statement

The data for the thesis is sourced from various websites, and the owner of the data is the respective websites from which it was gathered. No data is collected from human participants or animals. The data is freely available for download from these websites, implying implicit consent for its use. The acquisition channel is the internet, specifically through downloading CSV files. All images and figures in the thesis are created by the author, and there is no usage of images that require external consent.

The thesis does not involve acquiring new data from human or animal participants. The data used in the thesis consists of factual information, such as numbers and financial data, and is representative in nature. There are no known biases in the dataset. As the study does not involve human or animal participants, there is no specific de-identification regime associated with the data.

The thesis does not incorporate code from another source; all code is developed by the author. A comprehensive list including version numbers of all libraries and frameworks used in the study is provided (Chapter 3.5).

The tools used for checking spelling and grammar include Microsoft Word combined with Grammarly. No other tools or services were employed for paraphrasing or typesetting the given text.

1. Introduction: Problem Statement & Research Goal

This research project aims to compare the performance of Autoregressive Integrated Moving Average (ARIMA), Long Short-Term Memory (LSTM), and 1-Dimensional Convolutional Neural Network (1D-CNN) models in short-term stock trend prediction, integrating historical and technical indicators. Unlike many existing models that focus solely on historical or technical indicators, this study combines both categories. LSTM, recognized as the state-of-the-art, is evaluated alongside ARIMA and CNN. The research uses the NASDAQ100 as a case study and explores the impact of different input and output time horizons on short-term stock trend predictions. Additionally, the study analyzes model performance across various sectors within the NASDAQ100, providing a comprehensive evaluation. Through addressing these challenges, the research reevaluates the applicability and performance of ARIMA and CNN, even in the context of LSTM's dominance. The evaluation will focus on accuracy, as measures like recall, precision, and F1-score are deemed less relevant for stock trend prediction, where the primary concern is profitability.

In the past decade, the use of machine learning for stock price prediction has risen significantly, challenging the efficient market hypothesis, and emphasizing a data-driven approach. This thesis focuses on the domain of stock trend prediction, a field of significance in the financial world. While the dominant model for short-term stock prediction is LSTM, incorporating technical indicators (Namin & Namin, 2018; Balaji et al., 2018, Li & Lao, 2017; Yao et al., 2018; Nelson et al., 2017; Borovka & Tsiamas, 2019; Chen & Ge, 2019, Kobiela et al., 2022; Dong, 2022). This study brings a unique perspective by incorporating historical and technical indicators, diverse input and output timeframes, and assessing predictive accuracy across various sectors within the NASDAQ100—an aspect often overlooked in existing literature.this research introduces a novel dimension by integrating historical and technical indicators, input and output time horizons, and testing predictive accuracy for different sectors within the NASDAQ100 as these are all overlooked by the literature. ARIMA, known for historical stock data, may outperform LSTM in this extended context (Kobiela et al., 2022). Beyond ARIMA and LSTM, the study explores 1D-CNN in stock trend prediction in the extended context (Di Persio & Honchar, 2016; Gunduz et al., 2017). This comprehensive comparison expands the understanding of machine learning models in the dynamic field of stock price prediction.

Given this argumentation we arrive at the following problem statement: This study addresses the challenge of identifying the most effective forecasting model (ARIMA, LSTM, or 1D-CNN) for short-term stock trend prediction. It explores the impact of diverse input features, including historical and technical indicators, and examines how model performance varies across different input and output lengths in various prediction sectors.

Societally, this study addresses the issue of mitigating the high loss rates experienced by day traders (Barber et al., 2014). Applying data models to trading offers a data-driven solution that removes emotional decision-making, particularly beneficial for retail investors. Moreover, the comparison of these models equips day traders with the knowledge to make more informed decisions based on underlying patterns in financial markets, empowering them to improve their trading strategies. Scientifically, this study addresses gaps in existing literature by integrating overlooked technical and historical indicators. Their inclusion is justified by their potential to provide deeper insights into market dynamics, offering a holistic view of factors influencing stock trends. With a more holistic approach the models are expected to excel previous studies with relatively simple models. Exploring different input and output time horizons is expected to enhance predictive accuracy, capturing diverse market behaviors. This approach aims to advance understanding by demonstrating the relevance of these features in predictive modeling. The research uniquely analyzes optimal configurations across NASDAQ100 sectors, filling a gap in current literature and providing valuable insights into the generalizability and effectiveness of proposed configurations.

Based on the preceding information, the primary research question is as follows:

How does the accuracy differ of LSTM, ARIMA, and 1D-CNN models in forecasting short-term trends (uptrend/downtrend) in the NASDAQ100, considering the inclusion of historical, fundamental, and technical indicators in different time horizons and sectors?

To provide comprehensive support for this inquiry, the following sub-research questions are posed:

- How does the accuracy of ARIMA, LSTM and 1D-CNN models differ when diverse indicator types, encompassing historical and technical indicators, are integrated into the analysis?
- How does the variation in input time horizons, spanning from 2 days to 6 days, impact the predictive accuracy when contrasting ARIMA, LSTM, and 1D-CNN models?
- How is the accuracy of ARIMA, LSTM, and 1D-CNN models affected by adjustments in output time horizons, ranging from 1 day to 5 days?
- How does dividing forecasted stock price trends into specific sectors influence the predictive precision of the optimal model?

The thesis finds that in forecasting short-term price trends across financial markets, LSTM outperforms ARIMA and 1D-CNN models. Its versatility in capturing closing prices, historical trends, and technical indicators establishes LSTM's superiority. The optimal configuration involves an input length of 2 and an output length of 5 days using solely technical indicators, making LSTM the most reliable choice for financial trend forecasting. The optimal configuration generalizes to different sectors within the NASDAQ100.

2. Related Work

2.1 LSTM and short-term stock predictions

Long Short-Term Memory (LSTM) models, has shown promise in predicting nonlinear stock price movements. Namin and Namin (2018) found LSTM outperforming ARIMA with an average RMSE reduction of 87.445% (511.481 to 64.213). The robustness of LSTM was evident as the number of training epochs had no discernible impact on its performance. Balaji et al. (2018) corroborated LSTM's superiority over alternatives like GRU in BANKEX data predictions, with LSTM achieving an accuracy ranging from 61.38% to 59.05% for 1-step and 4-step predictions. Li & Liao (2017) demonstrated LSTM's precision (47.3% accuracy) in daily stock price predictions. Nelson et al. (2017) achieved 53%-55% accuracy for 15-minute interval predictions on Sao Paolo Stock Exchange stocks. Yao et al. (2018) found LSTM profitable for short-term trading strategies. Borovka and Tsiamas (2019) proposed an ensemble method with LSTM, enhancing accuracy in intraday stock forecasting. Chen and Ge (2019) highlighted the effectiveness of Attention LSTM (AttLSTM) over traditional LSTM in predicting Hong Kong stock movements, emphasizing the continuous refinement of deep learning techniques in financial forecasting. In research conducted by Nabipour et al. (2020), an average accuracy of 84.90% was achieved when predicting stock market trends using an LSTM model solely based on technical indicators. The study involved categorizing stocks into different groups. Another study by Nelson et al. (2017) focused on predicting the market trends of individual stocks using a similar LSTM model with only technical indicators, yielding an average accuracy of 54.26%. In Kobiela et al.'s (2022) study comparing ARIMA and LSTM models on logarithmized NASDAQ data, ARIMA outperformed LSTM in medium-short-term predictions, with a 3.4 times higher accuracy (MAPE of 1.64 vs. 5.52) in a 30-day window. However, LSTM slightly outperformed ARIMA in a 1-day window (MAPE 1.46 vs. 1.64). Kobiela et al. (2022) suggests exploring additional features to determine if LSTM can surpass ARIMA. Dong (2022) corroborates these findings in a study on Google's stock prices, where LSTM outperformed ARIMA in accuracy (RMSE 46.8 vs.

84.4) for short-term predictions. Future research aims to incorporate related stocks' closing prices for improved accuracy. These studies highlight the ongoing exploration of feature integration and model enhancements in the context of LSTM for stock trend prediction.

The highlighted studies emphasize LSTM models' effectiveness in addressing stock price time series nonlinearity, particularly showcasing their superiority over traditional methods like ARIMA and GRU (Namin and Namin, 2018; Balaji et al., 2018; Borovka and Tsiamas, 2019; Chen and Ge, 2019; Kobiela et al., 2022; Dong, 2022). However, these studies often lack a holistic view by neglecting historical and technical indicators and time horizons. This thesis addresses this limitation, incorporating multiple time horizons and historical indicators for improved accuracy. Unlike the literature, this thesis analyzes distinct NASDAQ100 sectors, enhancing predictive performance evaluation and aiming for more robust financial forecasting. While LSTM is renowned for short-term stock prediction, its applicability across diverse markets and susceptibility to external factors require further exploration. Notably, ANN models excel in long-term prediction, but their unsuitability for time series analysis and historical pattern assimilation led to their exclusion in this thesis. The omission of GRU and AttLSTM aligns with LSTM's acknowledgment as the contemporary benchmark (Sheth & Shah, 2023; Banerjee & Garg, 2022). In conclusion, while LSTM remains the state-of-the-art for short-term stock trend predictions, future research must explore their limitations, generalizability, and adaptability in diverse financial scenarios for a more comprehensive evaluation. Acknowledging these aspects will contribute to a more nuanced and well-rounded evaluation of LSTM models in the field of short-term stock price predictions.

2.2 ARIMA and short-term stock predictions

This thesis integrates ARIMA as a proposed model for short-term stock prediction, referencing extensive academic literature. Adewumi et al. (2014) demonstrated ARIMA's potential in forecasting stock exchanges, although specific accuracy metrics were not provided. Caginalp and Constantine (1995) observed ARIMA's outperformance in the financial domain. Yang et al. (2017) achieved a 2.4% error (RMSE 77.143) in predicting the Shanghai Composite Index. Kim and Sayama (2017) explored ARIMA and indicator combinations, reducing the S&P500 prediction error with relative strength (RMSE 5.018, MSE 20.27). Khashei and Hajirahmi (2019) dissected ARIMA into linear and non-linear components, with their hybrid ARIMA-ANN model slightly outperforming standalone ARIMA (RMSE 471.09 for ANN and 488.89 for ARIMA). They also noted the potential of combining ANN and ARIMA as a hybrid model for improved accuracy. Furthermore, in a research conducted by Uma Devi et al. (2013), an accuracy of 83.74% was achieved in predicting the price of the NIFTY-50 using only the closing price as input.

The highlighted studies, including Adewumi et al. (2014), Caginalp and Constantine (1995), Yang et al. (2017), Kim and Sayama (2017), and Khashei and Hajirahmi (2019), provide valuable insights into ARIMA models for short-term stock predictions. However, these studies often lack a comprehensive integration of technical and historical indicators, limiting the understanding of stock market dynamics. The absence of market sentiment analysis, recognized as influential (Gómez-Martínez et al., 2023; Johnson, 2023), and exploration of varied time horizons are notable gaps. Additionally, these studies predominantly focus on individual stocks or broad indices, overlooking the consideration of specific sectors in their evaluations. While Khashei and Hajirahmi (2019) consider ANN alongside ARIMA, comprehensive sentiment analysis remains overlooked. This research aims to address these limitations by systematically integrating technical indicators, considering market sentiment, and exploring varied time horizons to enhance ARIMA models' predictive power in short-term stock forecasting.

2.3 1D-CNN and short-term stock predictions

CNN, often overlooked in stock prediction, emerged as a dark horse, outperforming MLP and RNN models according to Di Persio and Honchar (2016). With an accuracy of 53.6% and MSE of 0.2491, CNN slightly surpassed multilayer perceptron (MLP) (52.1%) and recurrent neural network (RNN) (52.2%). Di Persio and Honchar emphasized feature preprocessing and achieved 56.9% accuracy using a blended ensemble of CNN, RNN, and W-CNN models. Gunduz et al. (2017) predicted hourly movements of 100 stocks on the Borsa Istanbul. Feature selection didn't significantly enhance logistics regression performance (MA F-Measure rate of 0.545). CNNs, focusing on feature correlations, outperformed logistic regression in both cases, with CNN-Corr achieving a higher F-Measure rate of 0.563. CNN is a viable option for short-term stock trend prediction.

The discussed studies illuminate the potential of CNN in stock prediction, yet this emerging field faces notable limitations. Primarily employed in image analysis and natural language processing, CNN's applications in stock prediction are relatively unexplored. Di Persio and Honchar's 2016 study shows CNN outperforming MLP and RNN models, but questions linger regarding the generalizability of findings across diverse feature inputs, input and output time horizons. The study underscores the importance of feature preprocessing, yet specific contributing features to CNN's outperformance lack exploration. While CNN holds promise, further research is needed to explore specific driving features, assess generalizability, and compare input features comprehensively for a nuanced evaluation of its role and limitations in stock price prediction (Di Persio & Honchar, 2016). It's worth noting that the consideration of different input/output lengths and diverse sectors has not been adequately addressed in research on short-term stock prediction using CNN.

2.4 Factors for predicting short-term stock trends

2.4.1 Historical indicators

These historical indicators, which encompass global economic factors, central bank decisions, commodity prices, and market sentiment, hold the potential to significantly contribute to the understanding and prediction of stock trend movements in this thesis. Historical stock prices, analyzed through technical analysis techniques, provide insights into future price movements by identifying recurring patterns such as moving averages, support and resistance levels, and chart patterns (Rutledge, 1986). The USD Index, representing major world currencies, reflects changes in global economic sentiment and impacts stock prices, including those in the NASDAQ100 (Bai & Koong, 2018). Federal Reserve interest rate decisions, with widespread consequences on borrowing costs, consumer spending, and investor preferences, also influence stock prices (Thorbecke, 2012). Furthermore, oil prices, having a ripple effect across sectors, are essential to monitor for informed investment decisions (Kang et al., 2015). Additionally, the CNN Fear and Greed Index, which gauges investor sentiment, indirectly influences stock prices (Gómez-Martínez et al., 2023; Johnson, 2023). Integrating these diverse historical indicators enhances the analytical framework, providing a more comprehensive understanding of the multifaceted factors influencing stock trends.

2.4.2 Technical indicators

This thesis incorporates essential technical indicators—Relative Strength Index (RSI), Moving Averages Convergence/Divergence (MACD), Average Directional Movement Index (ADX), Slow Stochastic Oscillator (STO %K and %D), and Exponential Moving Averages (EMA 21 & 26)—for a comprehensive exploration of short-term stock trend prediction. Renowned for distilling market information into quantifiable metrics, these indicators enhance the model's ability to discern nuanced patterns within stock price movements. RSI provides insights into recent price changes' magnitude, while MACD signals potential shifts in market trends. ADX indicates trend strength, and STO offers valuable information about closing prices in relation to their price range. Exponential Moving Averages (EMA 21 and 26) weigh recent price data more heavily, enhancing sensitivity to short-term fluctuations. This collection of technical indicators enriches the model's predictive capabilities, capturing diverse facets of market dynamics for a more robust short-term stock trend prediction framework.

The importance of technical indicators in short-term stock trend prediction is highlighted by their widespread application and proven efficacy in existing literature. Parida et al. (2023) used MACD and RSI for forecasting stock trends, and Piravechsakul et al. (2021) integrated STO %K and %D, MACD, and RSI, affirming their value in enhancing predictive models. Kaur et al. (2022) expanded by incorporating EMA, MACD, STO %K and %D, and RSI, further establishing the significance of these indicators in short-term trend analysis. Vincent et al. (2023) extended the usage to MACD, STO %K and %D, and RSI, corroborating their effectiveness. Liu (2021) focused on EMA and MACD, emphasizing their continued relevance. Notably, ADX lacks substantial representation in empirical studies, making its inclusion in this thesis as an additional technical indicator promising for contributing novel insights to the field of short-term stock trend prediction.

3. Methodology & Experimental setup

3.1 Experimental setup

This thesis utilizes numerical and daily features that influence the NASDAQ100. These variables are identified in existing literature as mentioned in the previous chapter and are therefore relevant to this thesis. The dataset spans from January 1, 2018, to December 30, 2022, encompassing a total of 1259 trading days. Non-trading days have been excluded from the dataset for analysis.

Feature	**Description**	**Source**
Opening Price	The price of an asset at the beginning the day	Yahoo Finance, 2023a
Closing Price	The price of an asset at the end of the day	Yahoo Finance, 2023a
Trading volume	The total number of shares or contracts traded during the day	Yahoo Finance, 2023a
USD index	A measure of the value of the U.S. dollar relative to a basket of other world currencies.	Yahoo Finance, 2023b
FED interest rate	The interest rate set by the Federal Reserve	ST. LOUIS FED, 2023
CNN Fear and Greed index	Measures investor sentiment in financial markets, indicating whether the market is driven by fear or greed based on various factors	CNN: Business, 2023
Oil Price (Crude)	The current value of a barrel of oil in the market, reflecting the supply and demand dynamics in the global oil industry	Investing.com, 2023
14-day RSI	A momentum oscillator measuring the speed and change of price movements	Calculated with the closing price of the NASDAQ 100

MACD	A trend-following momentum indicator that shows the relationship between two moving averages	Calculated with the closing price of the NASDAQ 100
ADX	A trend strength indicator used to assess the strength of a price trend	Calculated with the closing price of the NASDAQ 100
SO %K	Indicators measuring the momentum of price and identifying overbought or oversold conditions	Calculated with the closing price of the NASDAQ 100
SO %D	Indicators measuring the momentum of price and identifying overbought or oversold conditions	Calculated with the closing price of the NASDAQ 100
EMA 12	Moving averages that give more weight to recent prices, helping identify trends	Calculated with the closing price of the NASDAQ 100
EMA 26	Moving averages that give more weight to recent prices, helping identify trends	Calculated with the closing price of the NASDAQ 100

Table 1: All features, description, and their source

All formulas of the technical indicators can be found in appendix A.

Non-trading days, serving no purpose in predicting short-term stock trends, are discarded from the dataset. Most features remain unchanged during these days as assets cannot be traded in the open market. The initial 27 days are excluded, as they are mostly empty for technical indicators. ADX, taking 27 days to assess the strength of a price trend, contributes to the highest number of empty values. Consequently, the dataset effectively comprises 1232 trading days. Opening price, closing price, and trading volume are scaled using the Sci-kit MinMaxScaler (Pedregosa, et al., 2011) to create a stable, consistent, and optimized input for the models, enhancing their performance and convergence during training.

The thesis will leverage a combination of historical data and technical indicators, providing a comparative analysis against the prevailing state-of-the-art approach that relies solely on technical indicators using an LSTM model. Additionally, it will assess the performance in comparison to a baseline scenario using only the closing price, a secondary but widely used option. While not all historical and technical indicators in the dataset exhibit a strong correlation with the closing price, this is inconsequential. Existing literature supports the effectiveness and influential nature of all features considered, justifying their inclusion in the analysis. In the picture below, a notable observation is the presence of a correlation coefficient of precisely 1.00 between the opening and closing prices. This correlation arises from the inherent linkage between consecutive days, wherein the closing price of a particular day, such as day 1, aligns with the opening price of the subsequent day (day 2), establishing a direct interdependence between these variables.

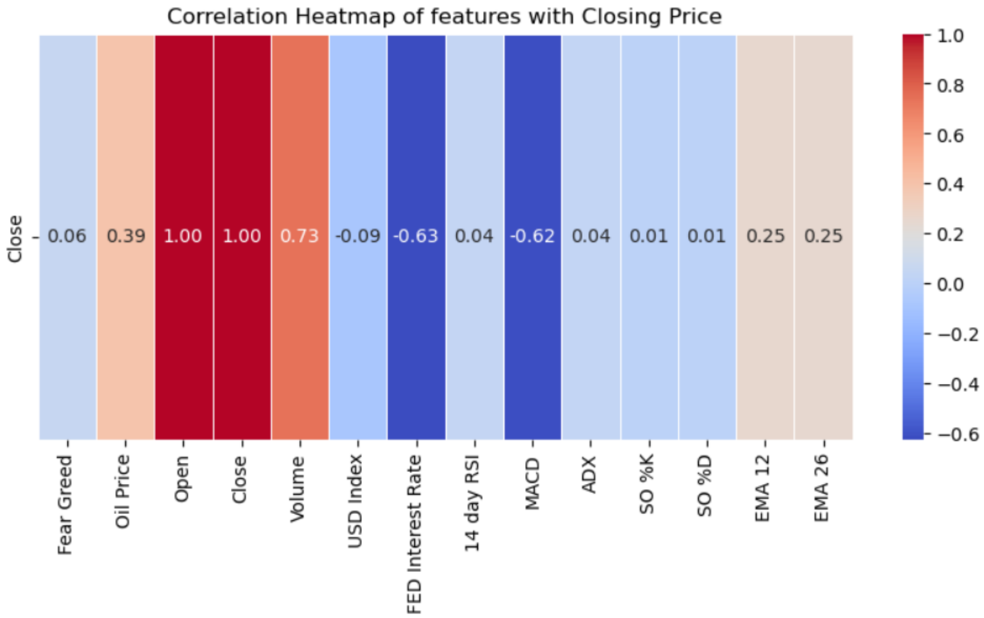

Image 1: Correlation heatmap of features with the closing price

The model employs input sequences of 2, 4, and 6 days using a rolling window approach for prediction. These lengths are chosen to closely align input with output, maximizing correlation. The rolling window ensures consideration of a continuous and overlapping series of historical data, providing a dynamic context for each prediction. The model predicts trends over 1, 3, and 5 days, with the rolling window capturing the evolving nature of input sequences for effective prediction. The models will predict either an 'uptrend' or a 'downtrend', serving as a conceivable measure for traders, as suggested by Hassanpour (2020). The visualization below illustrates the relationship between the lag period, represented on the x-axis for a rolling window spanning 1 to 350 days, and the corresponding average correlation with the closing price, depicted on the y-axis.

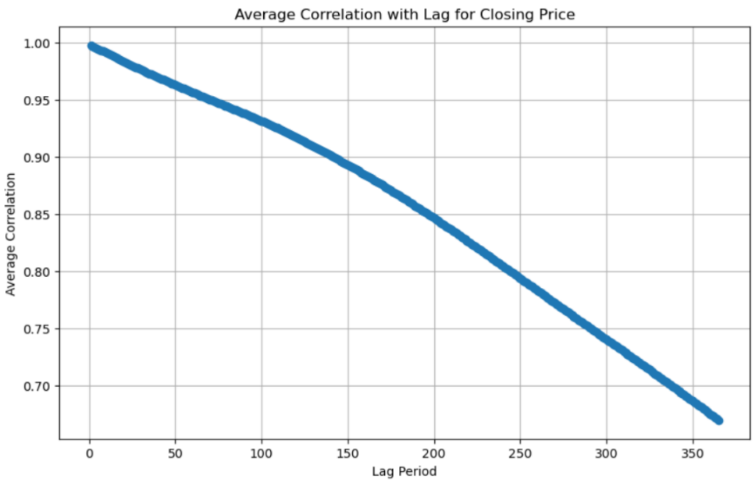

Image 2: average correlation with lag for the closing price

The top-performing model, identified through accuracy, will be systematically applied to diverse sectors within the NASDAQ100, aiming to assess its adaptability and effectiveness across varied market dynamics. This sector-wise analysis aims to uncover the model's robustness and generalization capability, contributing to a comprehensive understanding of its effectiveness under different conditions for broader financial forecasting purposes. Features such as opening price, closing price, and trading volume will be utilized across sectors, with technical indicators adjusted based on changing closing prices. The study focuses on representative samples of at least 3 stocks per sector, as identified by NASDAQ (2023), to manage data abundance. A full breakdown of which stocks are included in each sector and will be used in this thesis can be found in appendix B.

To assess the practical implications of this thesis, the model with the highest average test accuracy will undergo a simulation to evaluate its financial performance. The simulation involves executing buy or sell actions for the NASDAQ100 based on the model's predictions. Each simulation spans the entire period from the beginning to the end corresponding with the output sequence length, and this process will be repeated for every combination of input and output lengths using the best model.

The data in this thesis is split into three sets for robust model evaluation: a 70% training set for learning historical patterns, and a combined 30% validation-test set further divided into 15% for hyperparameter tuning (validation set) and 15% as a benchmark (test set). Random seeds ensure result reproducibility and transparency. To enhance reliability, each model undergoes three runs with random different seeds (23, 27, and 98 (RANDOM.ORG, 2023)), and their mean test accuracy is computed. This approach aims to mitigate biases and provide a comprehensive evaluation of model performance. Furthermore, a table with the minimum and maximum test accuracy values for each model along with their respective configurations will be presented. This approach facilitates the examination of whether the observed results stem from randomization effects or if there is consistency among different seed values.

3.2 LSTM

Long Short-Term Memory (LSTM) is a specialized type of recurrent neural network (RNN) designed for effective time series forecasting. This application of LSTM also stretches to the financial domain, and more specifically short-term stock prediction (Namin & Namin, 2018; Balaji et al., 2018, Li & Lao, 2017; Yao et al., 2018; Nelson et al., 2017; Borovka & Tsiamas, 2019; Chen & Ge, 2019, Kobiela et al., 2022; Dong, 2022).

The LSTM model architecture was defined using the Keras Sequential API (Chollet F. et al., 2023). The model consisted of a single LSTM layer with 64 units, configured to accept input sequences of varying lengths. The activation function for the LSTM layer was set to 'relu,' and a dropout rate was introduced to mitigate overfitting. A dense layer with a sigmoid activation function served as the output layer, reflecting the binary nature of the financial trend prediction task.

To enhance the performance of the LSTM model, a systematic hyperparameter tuning approach was implemented. The model underwent training and validation for different input sequence lengths (2, 4, and 6) through a comprehensive grid search strategy. Key hyperparameters explored included the optimizer choice ('adam' or 'sgd'), activation function ('relu' or 'tanh'), and dropout rate (0.0, 0.2, or 0.4). The grid search was implemented through scikit-learn's GridSearchCV (Pedregosa, et al., 2011), employing a three-fold cross-validation setup. The best-performing model, determined by the highest validation accuracy, was selected for predictions on the test set.

The dataset was split into training (70%), validation (15%), and test (15%) sets, ensuring a robust evaluation of the model's generalization capability. This division helps prevent overfitting by training the model on a substantial portion of the data, validating its performance on a separate set, and ultimately testing it on an independent dataset. The diverse input sequence lengths and hyperparameter combinations, coupled with a rigorous training-validation-test split, contribute to the model's robustness and its ability to generalize well to new, unseen data, promoting effective out-of-sample predictions.

The LSTM methodology chosen for short-term stock prediction ensures robust model comparison and generalization evaluation. The architecture includes a single layer of 64 units, and hyperparameter tuning systematically explores key parameters through grid search with three-fold cross-validation. The dataset is strategically split into training (70%), validation (15%), and test (15%) sets to prevent overfitting. This meticulous approach, including diverse sequence lengths and rigorous dataset splitting, enhances the LSTM model's robustness and generalization capability to previously unseen data.

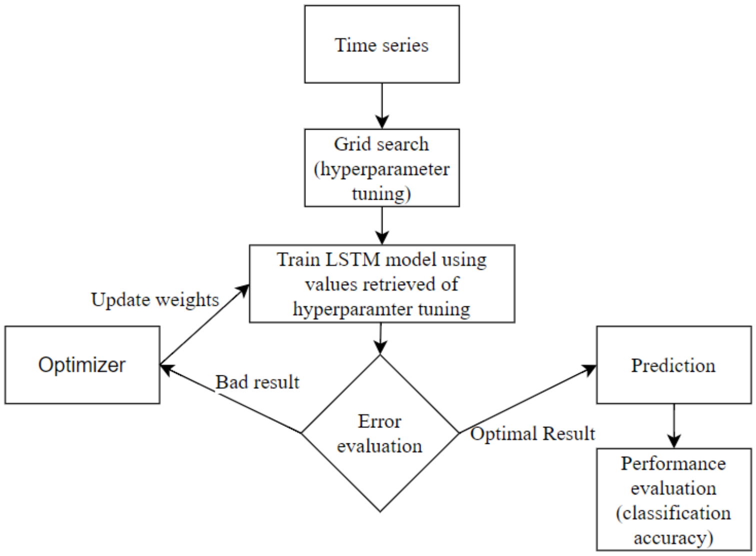

Image 3: Flowchart of the LSTM model

3.3 ARIMA

The ARIMA model is employed for time series prediction including time series predictions in the financial domain. ARIMA can be applied to short-term stock prediction (Adewumi et al., 2014; Caginalp & Constantine, 1995; Yang et al., 2017; Kim & Sayama, 2017; Khashei & Hajirahmi, 2019) ARIMA is a time series forecasting method. It combines autoregression (AR), differencing (I), and moving averages (MA) to model and predict the future values of a time series based on its historical patterns.

A hyperparameter tuning process is executed on the validation set to optimize the model's parameters, specifically the autoregressive order (p), differencing order (d), and moving average order (q). Setting the values of p and q to the input length and examining each number in the range of 5 for d. The best-tuned ARIMA model per input and output length is subsequently trained on the combined training and validation sets. Predictions are generated for both the validation and test sets, and model performance is assessed through classification accuracy.

The ARIMA model, employed for short-term stock prediction, demonstrates effectiveness and robustness in model comparison and generalization. Hyperparameter tuning on the validation set optimizes autoregressive, differencing, and moving average orders, with training on the combined dataset enhancing adaptability. Evaluation, focusing on classification accuracy, provides quantitative insights. Rigorous assessment of predictions includes calculating the p-value using different seeds and averaging the results, affirming ARIMA's reliability in capturing patterns and its generalization capability in short-term stock prediction scenarios.

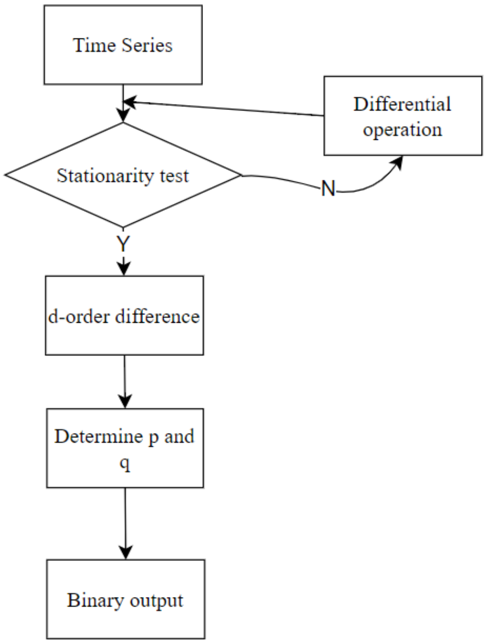

Image 4: Flowchart of the ARIMA model

3.4 1D-CNN

CNN is able to predict short-term stock trends and therefore is chosen as a model for this thesis (Di Persio & Honchar, 2016; Gunduz et al., 2017). The 1D-CNN model is constructed with three successive Conv1D layers, each associated with a distinct filter size. This architectural choice is pivotal in capturing hierarchical patterns within the temporal sequences of financial data, contributing to the model's capacity to discern nuanced trends.

The initial filter, with 32 units, captures low-level patterns, while subsequent filters (64 and 128 units) extract increasingly complex features. This hierarchical convolutional architecture discerns patterns of varying granularity, allowing the model to identify local and global dependencies within temporal sequences. As the CNN model runs efficiently, maxpooling is omitted, ensuring a reasonable timeframe for execution.

Notably, CNN is employed to discern binary trends across distinct temporal horizons (1 day, 3 days, and 5 days). The feature vectors are standardized, and a CNN model is instantiated with varying input day configurations (2, 4, and 6). Both contributing to robust out-of-sample model evaluation.

The thesis utilizes GridSearchCV (Pedregosa, et al., 2011) for extensive hyperparameter tuning. The hyperparameters explored include:

- Dropout rate: [0.2, 0.3, 0.4]
- Filter configurations: [[32, 64, 128], [64, 128, 256], [32, 64, 128, 256]]
- Kernel size: [3, 5]
- Dense layer units: [64, 128, 256]

The optimal hyperparameters, determined through cross-validated performance assessment on the validation set, guide the instantiation of an optimized CNN model, which is then evaluated on the validation, and test sets.

The 1D-CNN method for short-term stock trend prediction ensures robust model comparison and comprehensive evaluation beyond training data. Employing three Conv1D layers with varying filter sizes captures hierarchical patterns, allowing nuanced trend discernment. Stratified data division (70%/15%/15%) and consideration of temporal horizons (1 day, 3 days, and 5 days) contribute to thorough out-of-sample model evaluation. Hyperparameter tuning via GridSearchCV optimizes performance for dropout rate, filter configurations, kernel size, and dense layer units. Daily accuracy computation provides a nuanced understanding of the model's robustness and effective generalization to new, unseen data.

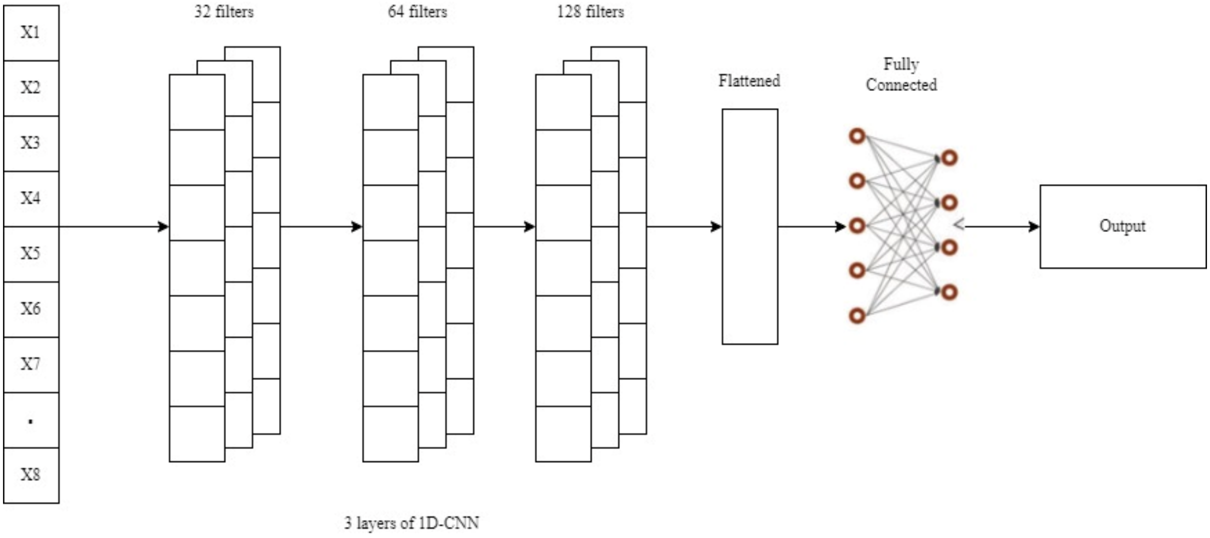

Image 5: flowchart of the CNN model

3.5 Evaluation and implementation

The thesis employs a comprehensive evaluation framework encompassing various metrics tailored to the tasks. Classification accuracy is a fundamental metric aligning with other studies, measuring the proportion of correctly predicted instances. While some studies use RMSE and MSE for stock price prediction, their application is beyond this study's scope due to a small input window size, potentially leading to inaccurate predictions. The inclusion of precision, recall, and F1 score supplements accuracy, providing a nuanced understanding of model performance and capturing strengths and limitations across domains. Additionally, the thesis thoughtfully contrasts machine learning models with a baseline for a thorough model comparison.

The baseline model represents the current state-of-the-art, utilizing an LSTM model with solely technical indicators. An alternative approach, close to the state-of-the-art, involves LSTM prediction solely with the closing price as a feature, which is also considered as a baseline. Additionally, predictions will be compared with a simple "naive" or "lagged" model, using the previous period's 'uptrend' or 'downtrend' to predict the next period, establishing a minimalistic baseline. The leading model identified in this thesis will assess various sectors within the NASDAQ100 to determine its generalizability across diverse industries. Finally, the model with the highest average accuracy will be used to simulate trading, providing insights into its practical implementation, considering the market's inherent upswings and downswings that may lead to potential losses for traders.

The thesis, developed in Python3 within JupyterLab, utilizes Python's versatility for data science and machine learning. TensorFlow (v2.14.0) is a key library employed for constructing and training neural network models, focusing on deep learning (Abadi et al., 2015). Other essential libraries include scikit-learn (v1.3.2) for machine learning algorithms (Pedregosa, et al., 2011), statsmodels (v0.14.0) for time series analysis (Seabold & Perktold, 2010), and seaborn/matplotlib (v0.12.0/v3.8.2) for data visualization (Hunter, 2007; Waskom, 2021). This comprehensive approach integrates deep learning with traditional statistical methods, contributing to a robust analytical framework.

4. Results

The results display test accuracies for all input and output lengths. Optimal hyperparameters per model can be found in appendix C. Due to space constraints, only optimal configuration confusion matrices for ARIMA, LSTM, and CNN are shown; others are in appendix D. Note that F1-score is reported in appendix E but not analyzed. In stock prediction, profitability is the focus, prioritizing accuracy over a balance between precision and recall, making F1-score less relevant to trading success. All test accuracies mentioned represent the mean accuracy of the three seeds described in the method. The full results are available in appendix E.

4.1 LSTM

The LSTM model's performance with historical and technical indicators ranges from 47.03% to 61.08%. However, it's crucial to highlight that there is a noticeable increase in accuracy for longer output lengths, particularly evident in the 5-day scenarios. The test accuracy, with an average of 54.76%, indicating that the model is capturing meaningful patterns but faces challenges in maintaining accuracy over shortened prediction horizons. Further investigation into improving performance for longer output lengths or exploring alternative model architectures may be considered.

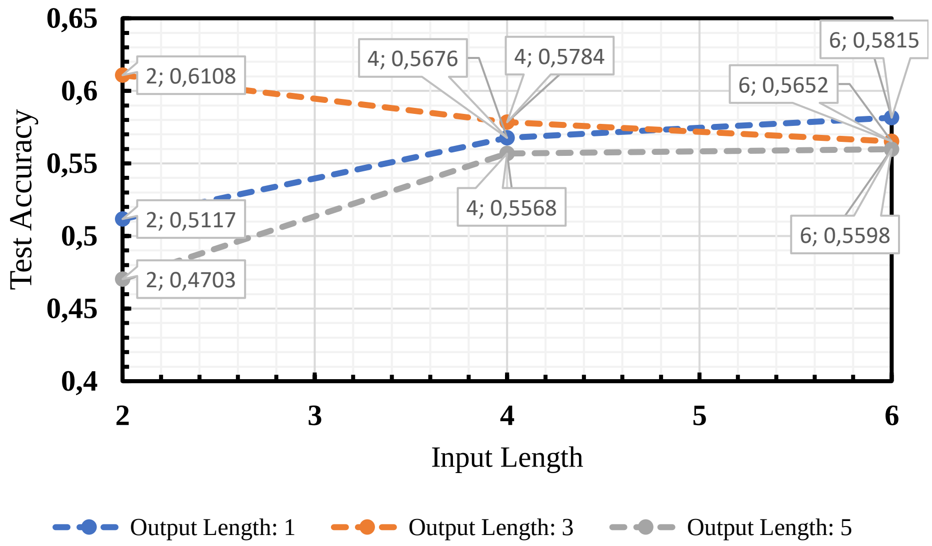

Image 5: Mean accuracy for different input and output lengths for LSTM with historical and technical indicators

The corresponding test accuracy for the technical indicators maintains a similar trend, averaging 66.93%. However, there's a notable incline in accuracy for longer output lengths, especially in 5-day scenarios. While the model demonstrates competence in capturing longer-term patterns, optimizing performance for shortened prediction horizons may be worth exploring through potential adjustments or alternative model architectures.

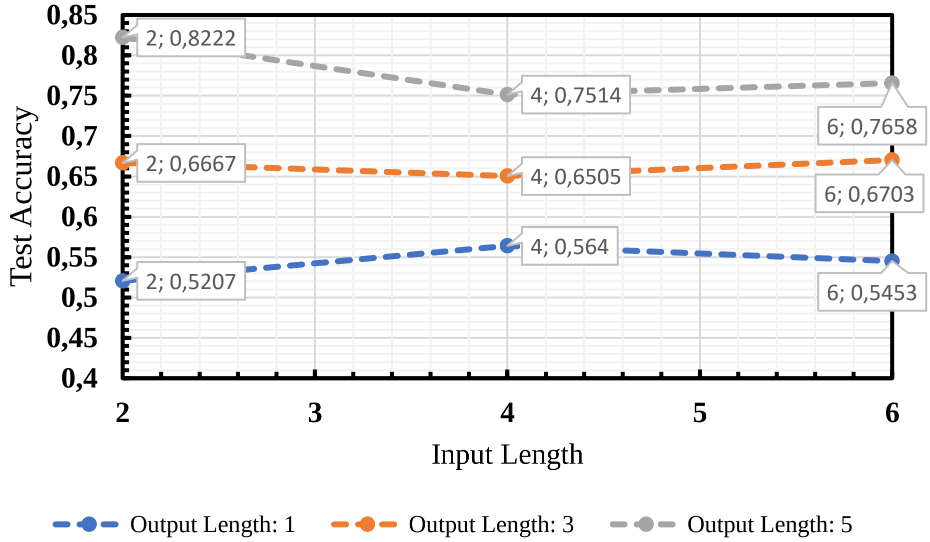

Image 6: Mean accuracy for different input and output lengths for LSTM technical indicators

The confusion matrix indicates that the best model (technical indicators, input 2 days, output 5 days) correctly predicted Uptrend 77.78% of the time and Downtrend 86.67% of the time. However, it mistakenly classified instances as Downtrend 22.22% of the time and Uptrend 13.33% of the time.

O/P	Uptrend	Downtrend
Uptrend	77.78%	22.22%
Downtrend	13.33%	86.67%

Table 2: Confusion matrix of the optimal configuration; technical indicators with 2 input days and 5 output days

For the LSTM with historical indicators the test accuracy presents a somewhat varied picture, averaging 55.32%. There is a steep decline in the accuracy for the input length 4 combined with output 1 and 3. The model's ability to generalize to unseen data is apparent, although there's room for improvement, particularly in preserving accuracy over longer output lengths. Exploring potential enhancements or alternative model architectures may offer avenues for further optimization.

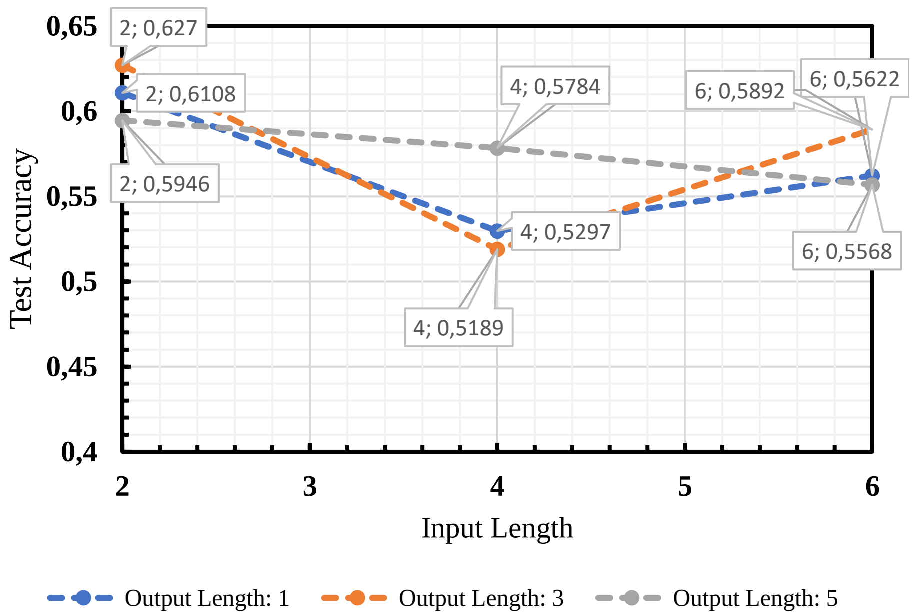

Image 7: Mean accuracy for different input and output lengths for LSTM with historical

The LSTM model with just the closing price as feature performs reasonably well when predicting stock trends based on closing prices compared to other features. There is a noticeable drop in accuracy for certain configurations, especially evident in 5-day output scenarios. This decline is reflected in the average test accuracy of 54.54%, suggesting that the model faces challenges in maintaining consistent performance, particularly over longer prediction horizons.

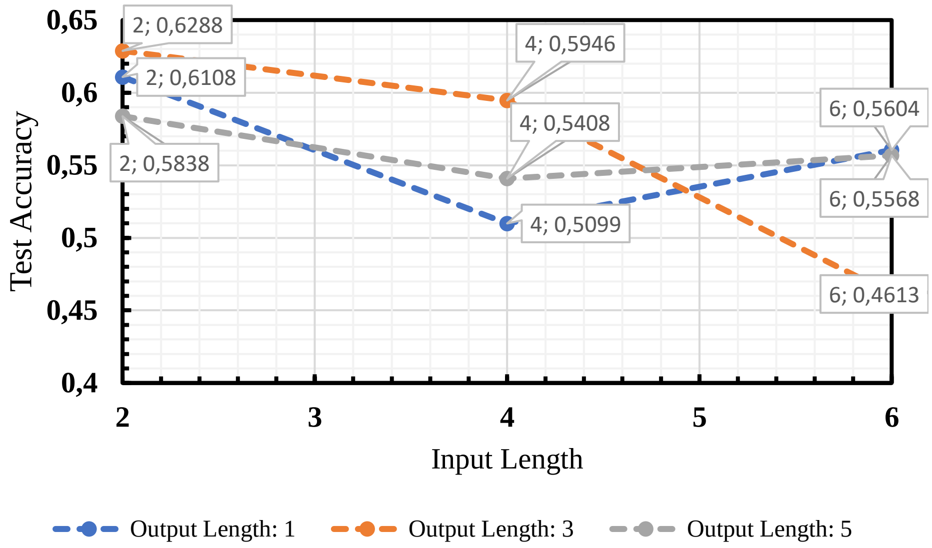

Image 8: Mean accuracy for different input and output lengths for LSTM with the closing price

LSTM models, leveraging technical indicators, consistently outperform counterparts with an average accuracy of 66.93%. Notably, reliance on closing prices alone yields lower accuracy, especially in longer projections, revealing challenges in sustained predictive power. Historical indicator-based models achieve an average accuracy of 55.32. The top LSTM model, with a 5-day output and 2-day input, emphasizes feature selection and sequence length's role in optimizing accuracy. In univariate setups, the LSTM achieves 55.78% average accuracy, excelling in 1-day projections (55.58%) and maintaining strong performance in 3 and 5-day forecasts (56.99% and 55.58%, respectively). These findings underscore LSTM models' robustness in financial trend forecasting, particularly when focusing on essential features like historical and technical indicators. Notably, the univariate setup outperforms in 1-day predictions and, in certain cases, specific input sequence lengths for 3-day forecasts.

Surprisingly, the number of epochs in the model has no impact on outcomes due to the random nature of each epoch, with no discernible progression in accuracy.

Model		Test Accuracy
LSTM: Historical and technical indicators (proposed model)		0.5476
LSTM: Technical indicators (baseline/state-of-the-art)		0.6619
Univariate setup (baseline)		0.5192

Table 3: Model comparison of the LSTM Performance

The table below presents the test accuracy outcomes for an LSTM model across various configurations and seed values. Notably, the results exhibit considerable variability in the performance of the model under different settings. For instance, when considering the 'Historical and technical' context with an output length of 1 and input lengths of 2, 4, and 6, the test accuracy ranges from 0.4649 to 0.5815. Furthermore, the standard deviations for these configurations are relatively low, indicating consistency across the three seeds. On the other hand, the 'Technical' context with an output length of 5 and input lengths of 2, 4, and 6 shows consistent maximum test accuracy values of 0.8054 across all seeds, with minimal standard deviations. In contrast, the 'Closing price' context exhibits substantial variability, especially for configurations with an input length of 4 and 6, where the test accuracy ranges from 0.2378 to 0.6324. This suggests that the choice of context and input length significantly influences the model's performance, and the impact of seed variation varies depending on the specific configuration. Further investigation may be warranted to understand the factors contributing to these observed patterns.

Indicators	Output Length	Input Length	Minimum test accuracy of seeds	Maximum test accuracy of seeds	Standard Deviation
Historical and technical	1	2	0.4649	0.5351	0.0406
Historical and technical	1	4	0.5189	0.5676	0.0281
Historical and technical	1	6	0.5815	0.5815	0.0000
Historical and technical	3	2	0.3892	0.6108	0.1280
Historical and technical	3	4	0.5784	0.5784	0.0000
Historical and technical	3	6	0.5652	0.5652	0.0000
Historical and technical	5	2	0.4703	0.5297	0.0343
Historical and technical	5	4	0.5568	0.5568	0.0000
Historical and technical	5	6	0.5598	0.5598	0.0000
Technical	1	2	0.4919	0.5622	0.0352
Technical	1	4	0.4649	0.5946	0.0667
Technical	1	6	0.5598	0.5815	0.0109
Technical	3	2	0.6865	0.7135	0.0136

Technical	3	4	0.6270	0.6703	0.0236
Technical	3	6	0.6467	0.6630	0.0094
Technical	5	2	0.7892	0.8054	0.0083
Technical	5	4	0.7351	0.7622	0.0143
Technical	5	6	0.8043	0.8478	0.0220
Historical	1	2	0.3892	0.6108	0.1280
Historical	1	4	0.5297	0.5297	0.0000
Historical	1	6	0.5622	0.5622	0.0000
Historical	3	2	0.3730	0.5656	0.1014
Historical	3	4	0.5676	0.5837	0.0093
Historical	3	6	0.5769	0.6054	0.0164
Historical	5	2	0.5027	0.5946	0.0531
Historical	5	4	0.5622	0.5784	0.0094
Historical	5	6	0.5568	0.5568	0.0000
Closing price	1	2	0.4865	0.6108	0.0718
Closing price	1	4	0.4811	0.5297	0.0281
Closing price	1	6	0.4919	0.5622	0.0406
Closing price	3	2	0.6270	0.6270	0.0000
Closing price	3	4	0.4324	0.5568	0.0689
Closing price	3	6	0.6054	0.6324	0.0156
Closing price	5	2	0.7027	0.7297	0.0156
Closing price	5	4	0.2378	0.5784	0.1966
Closing price	5	6	0.3514	0.4432	0.0531

Table 4: Standard deviation, minimum and maximum accuracy for different input and output lengths and indicator groups for LSTM

4.2 ARIMA

The graph below provides a comprehensive overview of the ARIMA model's performance across various combinations of output and input sequence lengths for different groups of indicators. The ARIMA model exhibited diverse accuracy levels across distinct input sequence lengths and output prediction scenarios, showcasing a noteworthy trend of increasing accuracy with longer output prediction periods. Notably, the highest test accuracy for historical and technical indicators was achieved in the 5-day scenario with a 4-day input sequence, 78.38%. On average, the model attained a test accuracy of 64.38%. The observed pattern of accuracy improvement with extended output prediction days suggests the model's effectiveness in capturing and forecasting trends within the historical and technical indicators.

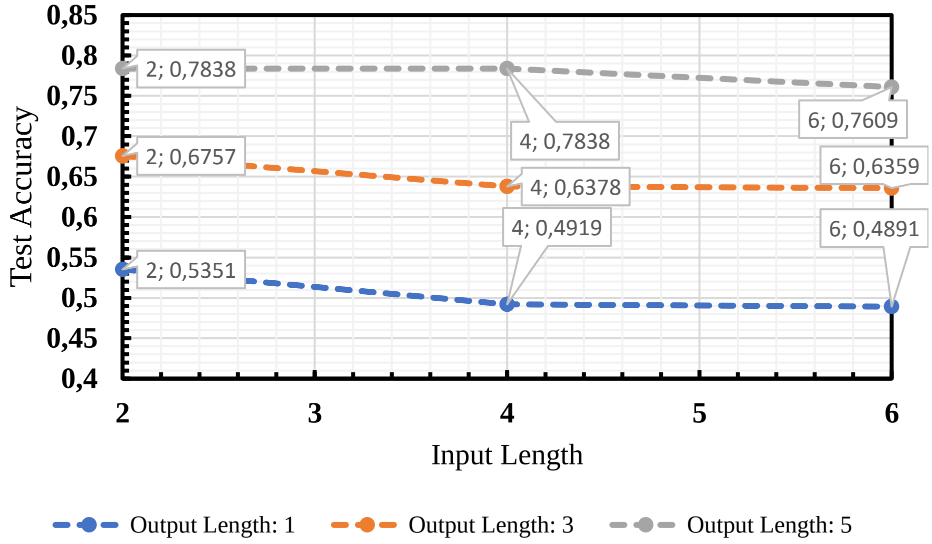

Image 9: Accuracy for different input and output lengths for ARIMA with historical and technical indicators

The ARIMA model using technical indicators also demonstrated varying accuracy levels based on different input sequence lengths and output prediction scenarios. Notably, similar to the historical and technical indicators case, there is an observable trend of increasing accuracy with longer output prediction periods. The model achieved its highest accuracy of 78.92% in the 5-day scenario with a 4-day input sequence. On average, the model achieved a test accuracy of 67.57%%. This consistency in the trend of improved accuracy with extended output prediction days suggests the model's robustness in leveraging technical indicators for forecasting.

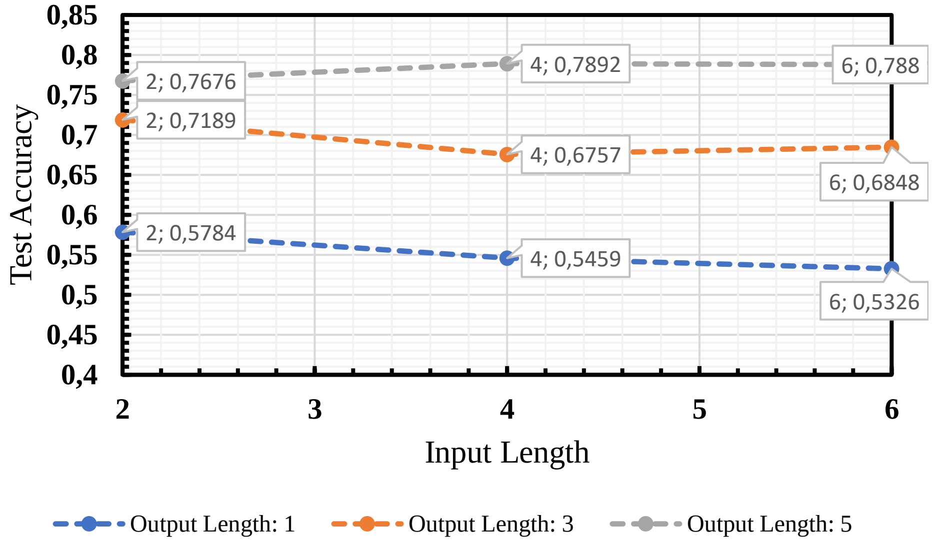

Image 10: Accuracy for different input and output lengths for ARIMA technical indicators

The confusion matrix indicates that the best model (input 4 days, output 5 days with technical indicators) correctly predicted Uptrend 83.08% of the time and Downtrend 73.44% of the time.

However, it mistakenly classified instances as Downtrend 26.56% of the time and Uptrend 16.92% of the time.

O/P	Uptrend	Downtrend
Uptrend	83.08%	16.92%
Downtrend	26.56%	73.44%

Table 5: Confusion matrix for the optimal configuration; technical indicators with 4 input days and 5 output days

Similar to the ARIMA models using historical and technical indicators, the ARIMA model relying solely on historical data also demonstrated a pattern of increasing accuracy with longer output prediction periods. The accuracy rates for this model range from 48.11% to 75.14%, and, on average, the model achieved a test accuracy of 63.42%. The highest accuracy of 75.14% was achieved in the 5-day output prediction scenario with a 6-day input sequence. This consistent improvement in accuracy for longer prediction periods suggests that the model benefits from a more extensive historical context when making predictions. Further analysis and comparison with other models would be essential to comprehensively assess the performance and predictive capabilities of this ARIMA model on the given dataset.

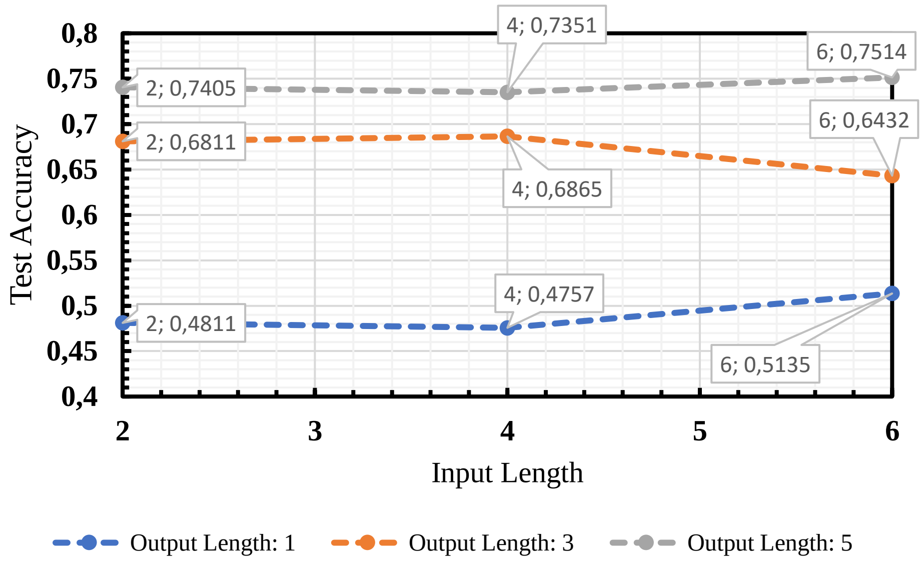

Image 11: Accuracy for different input and output lengths for ARIMA with historical indicators

For the ARIMA model based on the closing price, the trend of increasing accuracy with longer output prediction periods is also observed. The accuracy rates range from 42.16% to 74.59%, with an an average test accuracy of 58.38%. The highest accuracy of 74.59% is achieved in the 5-day output prediction scenario with a 6-day input sequence. This consistent improvement in accuracy for longer prediction periods suggests that the model benefits from a more extended historical context when making predictions based on closing prices. As with the other ARIMA models, a comprehensive evaluation would involve further analysis and comparison with alternative modeling approaches to gauge the model's effectiveness on the given dataset.

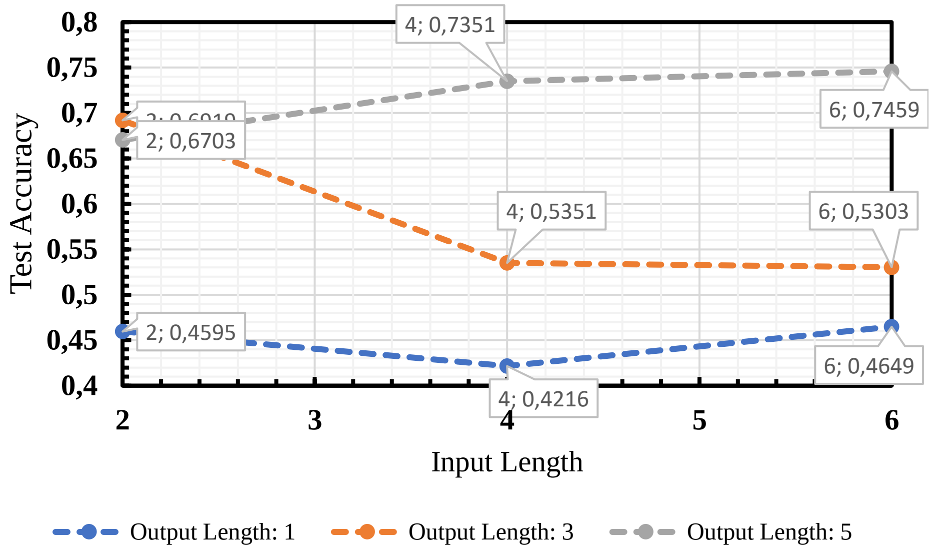

Image 12: Accuracy for different input and output lengths for ARIMA with the closing price

The ARIMA model, incorporating widely recognized technical indicators as input features, emerges as the top-performing approach in predicting stock price trends. Incorporating widely recognized technical indicators, the ARIMA model excels in predicting stock price trends. Using only the previous period's data, it achieves notable accuracies: 55.23% for 1-day, 69.31% for 3-day, and 78.16% for 5-day predictions, averaging to 67.57%. These findings affirm the efficacy of integrating technical indicators in ARIMA modeling for enhanced predictive performance in financial forecasting across various timeframes. The model's proficiency in capturing trends aligns with established literature recommendations, emphasizing the importance of relevant indicators.

Nevertheless, in the comparison between the proposed model of this thesis and the LSTM baseline, the proposed model falls short. However, it outperforms the baseline established by the univariate setup.

Model		Test Accuracy
ARIMA: Historical and technical indicators (proposed model)		0.6438
LSTM: Technical indicators (baseline/state-of-the-art)		0.6619
Univariate setup (baseline)		0.5191

Table 6: Model comparison of the ARIMA Performance

4.3 CNN

The image below provides a comprehensive overview of the CNN model's performance across various combinations of output and input sequence lengths for the different indicator groups. For historical and technical indicators, the highest accuracy is observed for a 5-day output length with a 6-day input length, reaching 71.58%. The average test accuracy across all configurations is 58.24%, showcasing the model's ability to capture underlying patterns in historical and technical data.

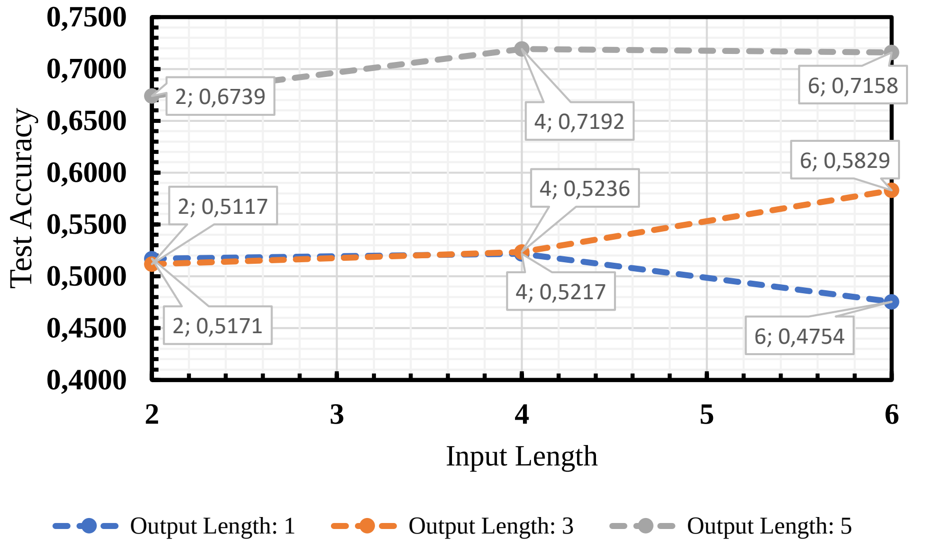

Image 13: Mean accuracy for different input and output lengths for CNN with historical and technical indicators

The confusion matrix indicates that the best model (input 6 days, output 5 days with historical and technical indicators) correctly predicted Uptrend 66.67% of the time and Downtrend 75.03% of the time. However, it mistakenly classified instances as Downtrend 24.97% of the time and Uptrend 33.33% of the time.

O/P	Uptrend	Downtrend
Uptrend	66.67%	33.33%
Downtrend	24.97%	75.03%

Table 7: Confusion matrix for the optimal configuration: Historical and technical indicators using 6 input days and 5 output days

For the technical indicators the model achieves the highest test accuracy of 73.69% for a 5-day output length and 2-day input length. However, it is essential to highlight a decline in accuracy for longer input lengths, indicating a potential challenge in capturing relevant technical patterns with extended historical information. These findings underscore the need for careful consideration of sequence length parameters to balance model complexity and predictive accuracy.

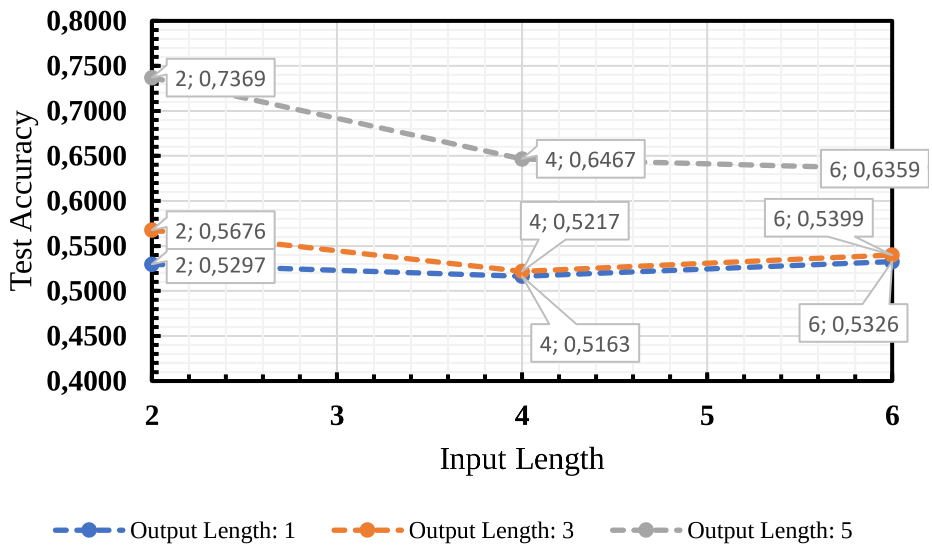

Image 14: Mean accuracy for different input and output lengths for CNN technical indicators

Noteworthy is the consistently high accuracy across different configurations for historical indicators, particularly achieving a peak validation accuracy of 66.67% for a 5-day output length and 2-day input length. The test accuracy averages slightly lower than models with using other indicators at 56.52%, indicating a potential challenge in generalizing predictions to unseen data. This suggests the need for further exploration to enhance the model's adaptability. Overall, the findings emphasize the effectiveness of the CNN model in leveraging historical data for trend prediction while emphasizing the importance of comprehensive evaluation metrics.

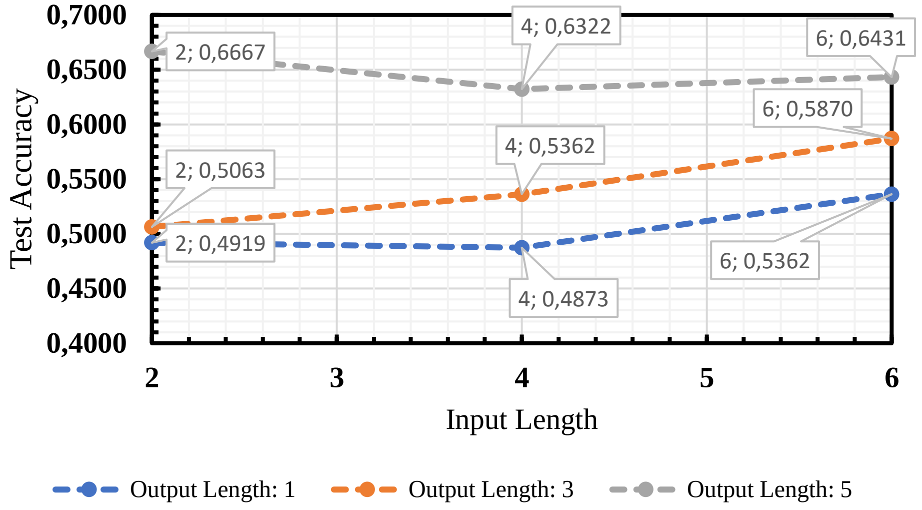

Image 15: Mean accuracy for different input and output lengths for CNN with historical

The model exhibits inconsistent accuracy across varied configurations for using solely the closing price, with notable peaks in test accuracy, such as 73.69% for a 5-day output length and 2-day input length. This indicates the model's proficiency in capturing patterns related to closing prices. The overall average test accuracy stands at 58.08%, reflecting the model's ability to generalize better to the test set.

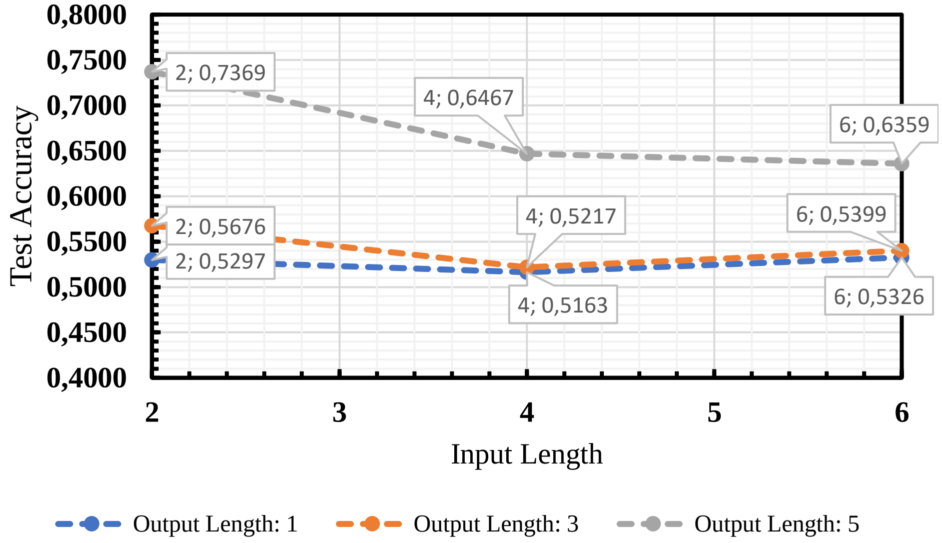

Image 16: Mean accuracy for different input and output lengths for CNN with the closing price

Surprisingly, the CNN model outperforms itself when using solely the closing price when compared to using technical indicators, showcasing an average test accuracy of 58.08%. These findings highlight the efficacy of leveraging convolutional neural networks, especially when focusing on closing prices, for robust financial trend forecasting. Notably, the univariate setup outperforms the CNN model in 1-day trend prediction and, in some instances, for specific input sequence lengths in 3-day trend prediction.

While the proposed model in this thesis does not outperform the LSTM baseline, it does surpass the baseline set by the univariate setup.

CNN: Historical and technical indicators (proposed model)	0.5823
LSTM: Technical indicators (baseline/state-of-the-art)	0.6693
Univariate setup (baseline)	0.5192

Table 8: Model comparison of the CNN Performance

The table below represents the performance metrics under various configurations, including output length, input length, and the associated minimum and maximum test accuracy of the different seeds. Observing the 'Historical and technical' context, the model exhibits varying test accuracy across different input lengths, with the minimum and maximum values ranging from 0.4590 to 0.7432. The standard deviations for these configurations are relatively moderate, suggesting a degree of consistency across seed variations. In the 'Technical' context, configurations with an input length of 5 consistently demonstrate high test accuracy, ranging from 0.6486 to 0.7213, with low standard deviations. Conversely, the 'Closing price' context reveals limited variability, particularly in configurations with an input length of 3 and 5, where the test accuracy remains constant across different seeds. Notably, for certain configurations, the standard deviations are minimal, indicating stable model performance across varied seeds. These findings underscore the sensitivity of CNN models to specific input and context parameters, influencing the variability in test accuracy outcomes. Further exploration may be necessary to elucidate the factors contributing to the observed patterns and their implications for model robustness.

Indicators	Output Length	Input Length	Minimum test accuracy of seeds	Maximum test accuracy of seeds	Standard Deviation
Historical and technical	1	2	0.4811	0.5351	0.0255
Historical and technical	1	4	0.5054	0.5489	0.0193
Historical and technical	1	6	0.4590	0.4918	0.0134
Historical and technical	3	2	0.4757	0.5459	0.0287
Historical and technical	3	4	0.5163	0.5380	0.0102

Historical and technical	3	6	0.5410	0.6066	0.0297
Historical and technical	5	2	0.6649	0.6811	0.0067
Historical and technical	5	4	0.7065	0.7337	0.0112
Historical and technical	5	6	0.6831	0.7432	0.0248
Technical	1	2	0.5081	0.5568	0.0202
Technical	1	4	0.4348	0.4837	0.0203
Technical	1	6	0.4317	0.5301	0.0402
Technical	3	2	0.4811	0.5459	0.0270
Technical	3	4	0.4728	0.4837	0.0044
Technical	3	6	0.4973	0.5410	0.0178
Technical	5	2	0.6486	0.6919	0.0192
Technical	5	4	0.6902	0.7065	0.0068
Technical	5	6	0.6721	0.7213	0.0201
Historical	1	2	0.4757	0.5027	0.0117
Historical	1	4	0.4728	0.5000	0.0112
Historical	1	6	0.5217	0.5489	0.0112
Historical	3	2	0.4973	0.5135	0.0067
Historical	3	4	0.5326	0.5380	0.0026
Historical	3	6	0.5815	0.5924	0.0044
Historical	5	2	0.6541	0.6757	0.0092
Historical	5	4	0.6141	0.6413	0.0128
Historical	5	6	0.6359	0.6467	0.0051
Closing price	1	2	0.5297	0.5297	0.0000
Closing price	1	4	0.5109	0.5217	0.0044
Closing price	1	6	0.5109	0.5489	0.0160
Closing price	3	2	0.5676	0.5676	0.0000
Closing price	3	4	0.5217	0.5217	0.0000
Closing price	3	6	0.5217	0.5489	0.0128
Closing price	5	2	0.7297	0.7459	0.0067
Closing price	5	4	0.6250	0.6685	0.0177

Closing price	5	6	0.6250	0.6522	0.0117

Table 8: Standard deviation, minimum and maximum accuracy for different input and output lengths and indicator groups for CNN

4.4 Sectors within NASDAQ

Testing the best model, LSTM with technical indicators, across diverse NASDAQ sectors provides insightful findings. In Consumer Discretionary and Consumer Staples, the model shows moderate performance, beating the univariate baseline with validation accuracies of 57.13% and 53.17%, and test accuracies of 54.11% and 51.37%, respectively. In Healthcare, Industrials, Technology, and Telecommunications, the LSTM model exhibits strong performance, with accuracies from 65.62% to 66.96%, showcasing its ability to generalize well across industries. This emphasizes the model's adeptness at forecasting trends in specific sectors. Overall, the LSTM model consistently performs well across sectors, highlighting its potential for broad applications in financial trend forecasting within the NASDAQ100.

Sector	Average Test Accuracy
Consumer Discretionary	0.5411
Consumer Staples	0.5137
Healthcare	0.6562
Industrials	0.6733
Technology	0.6695
Telecommunications	0.6588
Utilities	0.6643

Table 9: Results for different sectors of the NASDAQ100 using LSTM with technical indicators

4.5 Trading simulation

In the context of financial performance evaluation, an LSTM model, characterized by a 2-day input length and a 5-day output length, incorporating technical indicators, is subjected to a simulation. This model configuration has been selected due to its demonstrated highest accuracy in predictive capabilities. Notably, transaction costs are omitted from consideration to simplify the analytical framework. The primary objective of the LSTM model is to forecast the closing price of the NASDAQ, with an intentional temporal misalignment in subsequent predictions to assess the model's sustained predictive efficacy.

The dataset is partitioned into a training set (70%), a validation set (15%), and a prediction set (15%). It is imperative to emphasize that the LSTM model generates binary predictions, wherein a value of '1' signifies an anticipated uptrend, while '0' indicates a predicted downtrend. These binary predictions serve as the basis for trading signals, thereby forming the basis for subsequent analysis and assessment of the model's effectiveness in capturing market trends. This research framework allows for a comprehensive examination of the LSTM model's performance under diverse conditions and temporal sequences, contributing valuable insights to the field of financial prediction and modeling.

The trading simulation operates daily, initiating a trade each day with a fixed amount of $200. The direction of the trade, whether it is a buy or sell, is determined based on the model's prediction. The profit or loss of the trade is adjusted daily in accordance with the closing price changes. Trades initiated on day 1 will be cashed out on the 5th day. Trades initiated on day 2 will be cashed out on the 6th day, and so on. This aligns with the chosen output length.

The starting trading balance is set at $1000, and the simulation does not incorporate any leverage. This approach ensures a feasible trading simulation, addressing concerns related to randomization and maintaining the correct temporal ordering of events.

In addition to the LSTM model, two baseline scenarios are considered for comparison. The first baseline involves using 'true labels' as predictions, aiming to maximize profit. The second baseline employs the previous day's outcome as the prediction for the subsequent 5 days, providing a benchmark for comparison against the LSTM model's predictive capabilities. These baselines serve as essential references to gauge the relative success of the LSTM model in the context of the financial simulation.

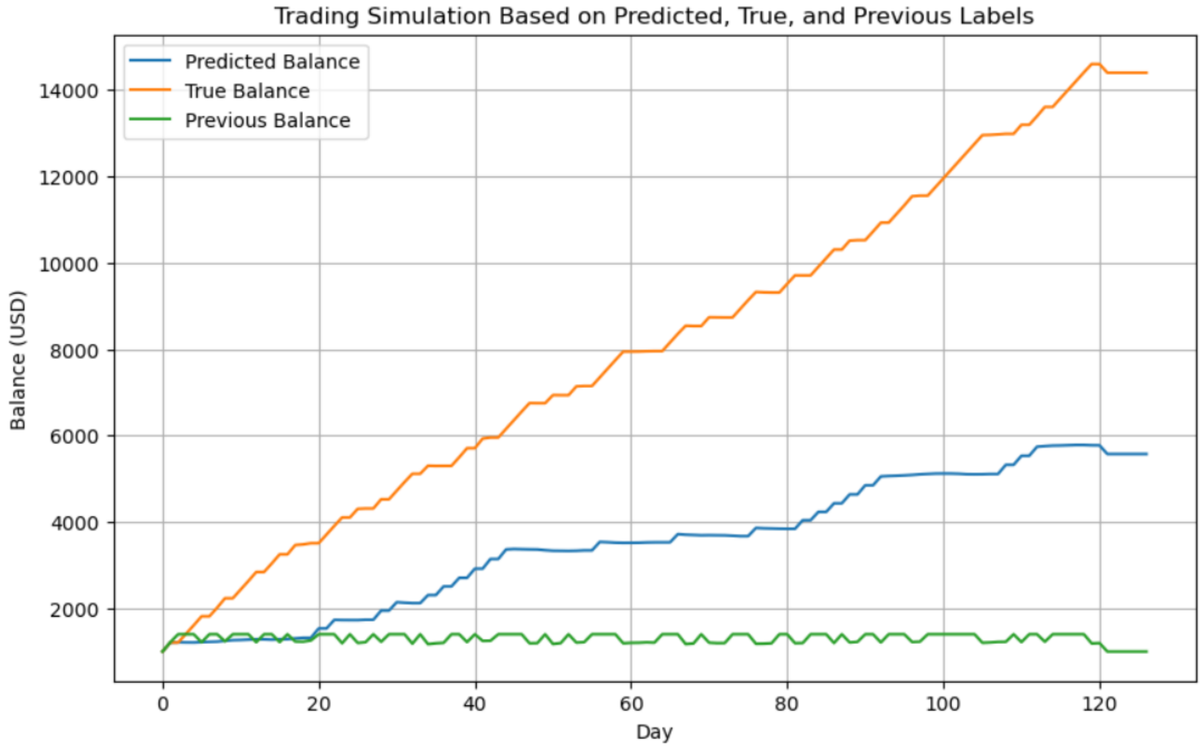

Image 17: Visualization of the trading simulation using LSTM with technical indicators, true labels, and previous labels

Ending Balance Based on Predicted Labels at Day 120: 5772.87 USD
Ending Balance Based on True Labels at Day 120: 14598.32 USD
Ending Balance Based on Previous Labels at Day 120: 1196.52 USD

To see all the trades in detail, including their individual profit and loss. Please refer to appendix F.

4.6 Comparison of models

Models using ARIMA, LSTM, and CNN with historical and technical indicators fall short of the state-of-the-art LSTM model relying solely on technical indicators. This emphasizes the supremacy of a focused approach, where LSTM excels with a streamlined set of features. Findings underscore the pivotal role of feature selection, suggesting that the LSTM model, leveraging impactful technical indicators, outperforms more complex counterparts. In financial time series forecasting, a strategic balance between model complexity and feature relevance is crucial for optimal predictive performance. The table below presents test accuracies averaged across all input and output lengths for the best model per ARIMA, LSTM, and CNN, supporting the statements above.

Model	Features	Average test accuracy

LSTM	Technical indicators (state-of-the-art)	0.6693
ARIMA	Technical indicators	0.6665
CNN	Historical and technical indicators	0.5824

Table 10: Accuracy comparison of the best performing features per model

The LSTM model consistently outperforms ARIMA and CNN models in financial trend forecasting, demonstrating robust performance across various configurations. It excels in predicting closing prices, historical trends, and technical indicators, confirming its state-of-the-art status in the domain. Versatile across input sequence lengths and output horizons, the LSTM model proves promising for capturing both short-term and long-term trends. Statistically significant results, higher average test accuracy, and an optimal configuration of 2 input and 5 output days reinforce the LSTM model's reliability and flexibility in financial time series forecasting, making it a top choice.

5. Discussion

This thesis aims to assess ARIMA, LSTM, and CNN models in financial time series forecasting, incorporating historical and technical indicators. Contrary to expectations, the outcomes consistently demonstrate the superior performance of the LSTM model, leveraging only technical indicators. The LSTM model excels in predicting closing prices, historical trends, and technical indicators, showcasing robust performance compared to ARIMA and CNN. The thesis identifies an optimal configuration for the LSTM model with an input length of 2 and an output length of 5 days. Furthermore, the LSTM model exhibits generalization across all individual sectors within the NASDAQ100. These findings underscore the dominance of the LSTM model in financial trend forecasting, emphasizing the importance of considering various input and output time horizons for a comprehensive understanding of its predictive performance.

This thesis advances understanding of LSTM models in stock price prediction, building on insights from studies like Namin and Namin (2018) and Balaji et al. (2018). The findings underscore LSTM's robustness, especially in short-term predictions, addressing a literature gap focused on shorter horizons. While supporting LSTM's effectiveness for short-term trading, our model falls short of Nabipour et al.'s (2020) 84.90% accuracy, possibly due to differences in technical indicators and their binary encoding approach. Extending Nelson et al.'s (2017) work, this study showcases LSTM's adaptability in various financial configurations and introduces a distinctive perspective by assessing LSTM models across diverse sectors within the NASDAQ100, contributing significantly to a literature landscape primarily focused on general accuracy comparisons.

Scientifically, the thesis advances stock forecasting by integrating technical and historical indicators into LSTM modeling, offering a comprehensive perspective on feature relevance and predictive accuracy. Building upon established studies (Namin and Namin, 2018; Balaji et al., 2018; Nabipour et al., 2020), this approach extends the analysis to enhance the understanding of temporal dynamics in predictive models. Societally, the thesis addresses the challenge of high loss rates for day traders, aiming to mitigate emotional decision-making in financial trading (Barber et al., 2014). By employing advanced data-driven models like LSTM and ARIMA, it provides a practical solution empowering retail investors to make more informed decisions and improve trading strategies. Furthermore, the analysis of LSTM, ARIMA, and 1D-CNN models on different segments within the NASDAQ100 fills a notable gap in the current literature, offering up-to-date insights into their comparative performance for both researchers and practitioners in the evolving landscape of financial markets.

6. Limitations and Future Research

An identified limitation in the thesis is the overfitting in many models due to the short input lengths used. Overfitting occurs when a model becomes too tailored to the training data, potentially capturing noise rather than genuine patterns. To address this, the thesis implemented standard practices such as cross-validation and careful parameter tuning. Future research could explore more advanced regularization techniques or ensemble methods to further mitigate overfitting risks associated with short input lengths. Another consideration involves the constraints of the historical stock data used. The dynamic nature of financial markets introduces uncertainties, and unexpected events or market anomalies could impact model performance. Future work could enhance robustness by incorporating a more extensive and diverse dataset, potentially integrating market shocks influencing financial markets. The thesis is constrained by specific input and output lengths, acknowledging that numerous alternative configurations exist for short-term stock trend prediction. The limitations inherent in the chosen lengths underscore the potential for further exploration and experimentation with diverse input and output configurations to enhance the depth of the analysis.

Future research should focus on identifying the specific technical indicators that consistently contribute to accurate predictions, guiding feature selection for enhanced model performance. A systematic analysis of various indicators and their impact on forecasting accuracy is crucial for valuable insights. Investigating the optimal output period is necessary for refining model configurations, with future work systematically exploring the point where prediction accuracy stabilizes or degrades, informing the selection of appropriate prediction horizons. Addressing overfitting concerns, especially with short input lengths, remains a focus, and exploring advanced techniques like regularization methods or ensemble approaches could improve model generalization and reduce overfitting risks. Beyond LSTM, ARIMA, and 1D-CNN models, future research could explore alternative architectures, including emerging models or hybrid approaches, offering novel perspectives on financial time series forecasting. Additionally, extending the thesis to real-time applications is crucial for practical implications, developing models that can adapt to evolving market conditions and incorporate up-to-the-minute data, bridging the gap between theoretical insights and actionable strategies for traders and investors.

7. Conclusion

The comparative analysis of ARIMA, LSTM, and 1D-CNN models in forecasting short-term price trends reveals LSTM's supremacy, leveraging only technical indicators for optimal predictive performance.

How does the accuracy of ARIMA, LSTM and 1D-CNN models differ when diverse indicator types, encompassing historical and technical indicators, are integrated into the analysis?
The thesis demonstrates that when diverse indicator types, encompassing both historical and technical indicators, are integrated into the analysis, the LSTM model consistently outperforms ARIMA and 1D-CNN. Contrary to expectations, LSTM leveraging only technical indicators exhibits superior predictive accuracy in forecasting financial time series, particularly in predicting closing prices, historical trends, and technical indicators.

How does the variation in input time horizons, spanning from 2 days to 6 days, impact the predictive accuracy when contrasting ARIMA, LSTM, and 1D-CNN models?
The variation in input time horizons, spanning from 2 days to 6 days, is systematically evaluated for ARIMA, LSTM, and 1D-CNN models. The findings underscore the importance of considering different input time horizons, revealing an optimal configuration for the LSTM model with an input

length of 2 days. This variation significantly influences the predictive accuracy of the models, further emphasizing the adaptability and robustness of LSTM across diverse time horizons.

How is the accuracy of ARIMA, LSTM, and 1D-CNN models affected by adjustments in output time horizons, ranging from 1 day to 5 days?
Adjustments in output time horizons, ranging from 1 day to 5 days, are investigated for ARIMA, LSTM, and 1D-CNN models. The thesis identifies that the LSTM model maintains its superior performance across various output time horizons, showcasing its efficacy in short-term predictions. This analysis provides insights into the model's ability to adapt and generate accurate forecasts for different temporal intervals.

How does dividing forecasted stock price trends into specific sectors influence the predictive precision of the optimal model?
The accuracy of ARIMA, LSTM, and 1D-CNN models is assessed concerning the division of forecasted stock price trends into specific sectors within the NASDAQ100. The study reveals that the LSTM model exhibits generalization across all individual sectors, reinforcing its dominance in financial trend forecasting. This sector-specific analysis introduces a distinctive perspective, contributing significantly to the literature landscape primarily focused on general accuracy comparisons.

The study emphasizes the balance between model simplicity and feature relevance, providing valuable guidance for practitioners and researchers. The identified optimal configuration and improved predictive capabilities offer practical guidelines for forecasting models in dynamic financial markets, reaffirming the LSTM model's supremacy with a focus on technical indicators.

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Appendix

Appendix A Technical indicator formulas

14-day RSI: $RS = \frac{Average\ 14\ day\ gain}{Average\ 14\ day\ loss}\ RSI = 100 - (\frac{100}{1+RS})$

MACD: $MACD = EMA(12) - EMA(26)$

ADX:

$$+\text{DI} = \left(\frac{\text{Smoothed } +\text{DM}}{\text{ATR}}\right) \times 100$$

$$\text{-DI} = \left(\frac{\text{Smoothed -DM}}{\text{ATR}}\right) \times 100$$

$$\text{DX} = \left(\frac{|+\text{DI} - \text{-DI}|}{|+\text{DI} + \text{-DI}|}\right) \times 100$$

$$\text{ADX} = \frac{(\text{Prior ADX} \times 13) + \text{Current ADX}}{14}$$

where:

+DM (Directional Movement) = Current High − PH

PH = Previous High

-DM = Previous Low − Current Low

Smoothed +/-DM = $\sum_{t=1}^{14} \text{DM} - \left(\frac{\sum_{t=1}^{14} \text{DM}}{14}\right) + \text{CDM}$

CDM = Current DM

ATR = Average True Range

SO %K: $SO\ \%K = \frac{Current\ close - Lowest\ Low}{Highest\ high - Lowest\ Low} * 100$
SO %D: 3-period SMA of %K

EMA 12: $EMA(12) = ((CMP * SF) + (EMA(p) * (1 - SF)))$
EMA 26: $EMA(26) = ((CMP * SF) + (EMA(p) * (1 - SF)))$

where:

CMP = Current market price
SF = Smoothing factor
EMA(p) = Previous EMA

Appendix B NASDAQ sectors

Sector	Stocks
Consumer discretionary	Amazon (AMZN). Starbucks (SBUX). Verisk Analytics (VRSK). Marriot International (MAR). Dollar Tree (DLTR)
Consumer staples	Monster Beverage (MNST). Kraft Heinz (KHC). PepsiCo (PEP). Amgen (AMGN). Mondelez International (MDLZ)
Health Care	Gilead Sciences (GILD). AstraZeneca (AZN). Seagen (SGEN). IDEXX Laboratories (IDXX)

Industrials	CSX corporation (CSX). Honeywell international (HON). Baker Hughes (BKR)
Technology	Autodesk (ADSK). Apple (AAPL). Alphabet incorporated (GOOGL). Microchip technology incorporated (MCHP)
Telecommunications	T-Mobile US (TMUS). Charter communications (CHTR). American Electric Power Company (AEP)
Utilities	Xcel Energy (XEL). Exelon corporation (EXC)

All data has been obtained from Yahoo Finance (2023).

Appendix C hyperparameters per model

LSTM:

Historical and technical indicators		
Output length	Input Length	Optimal hyperparameters (Activation. dropout rate. optimizer)
1	2	(Relu. 0.0. SGD)
1	4	(Relu. 0.2. Adam)
1	6	(tanh. 0.0. SGD)
3	2	(Relu. 0.2. SGD)
3	4	(tanh. 0.0. Adam)
3	6	(tanh. 0.0. Adam)
5	2	(Relu. 0.0. SGD)
5	4	(tanh. 0.0. Adam)
5	6	(tanh. 0.0. Adam)

Historical indicators		
Output length	Input Length	Optimal hyperparameters (Activation. dropout rate. optimizer)
1	2	(tanh. 0.0. Adam)
1	4	(tanh. 0.0. Adam)
1	6	(tanh. 0.0. SGD)
3	2	(Relu. 0.4. SGD)
3	4	(tanh. 0.0. Adam)
3	6	(tanh. 0.0. Adam)
5	2	(Relu. 0.4. Adam)
5	4	(Relu. 0.2. Adam)
5	6	(tanh. 0.0. Adam)

Technical indicators	

Output length	Input Length	Optimal hyperparameters (Activation. dropout rate. optimizer)
1	2	(tanh. 0.0. Adam)
1	4	(Relu. 0.2. Adam)
1	6	(tanh. 0.0. SGD)
3	2	(tanh. 0.2. Adam)
3	4	(tanh. 0.0. Adam)
3	6	(tanh. 0.2. Adam)
5	2	(tanh. 0.0. Adam)
5	4	(tanh. 0.2. Adam)
5	6	(tanh. 0.4. Adam)

Closing Price		
Output length	Input Length	Optimal hyperparameters (Activation. dropout rate. optimizer)
1	2	(Relu. 0.0. SGD)
1	4	(tanh. 0.0. Adam)
1	6	(tanh. 0.0. Adam)
3	2	(Relu. 0.0. SGD)
3	4	(Relu. 0.2. Adam)
3	6	(tanh. 0.0. Adam)
5	2	(Relu. 0.4. Adam)
5	4	(Relu. 0.0. SGD)
5	6	(Relu. 0.2. SGD)

ARIMA:

Historical and technical indicators		
Output length	Input Length	Optimal hyperparameters (p.d.q)
1	2	(0. 0. 0)
1	4	(1. 0. 2)
1	6	(2. 0. 3)
3	2	(0. 0. 0)
3	4	(0. 0. 0)
3	6	(0. 0. 0)
5	2	(0. 0. 0)
5	4	(0. 0. 0)
5	6	(0. 0. 3)

Historical indicators		
Output length	Input Length	Optimal hyperparameters (p.d.q)
1	2	(0. 0. 0)
1	4	(2. 0. 2)

1	6	(0. 0. 0)
3	2	(0. 0. 0)
3	4	(0. 0. 0)
3	6	(0. 0. 0)
5	2	(0. 0. 0)
5	4	(0. 0. 0)
5	6	(0. 0. 0)

Technical indicators		
Output length	Input Length	Optimal hyperparameters (p.d.q)
1	2	(1. 0. 1)
1	4	(3. 0. 3)
1	6	(4. 0. 3)
3	2	(0. 0. 1)
3	4	(0. 0. 1)
3	6	(0. 0. 1)
5	2	(0. 0. 1)
5	4	(2. 0. 2)
5	6	(0. 0. 1)

Closing Price		
Output length	Input Length	Optimal hyperparameters (p.d.q)
1	2	(0. 0. 0)
1	4	(0. 0. 0)
1	6	(1. 1. 3)
3	2	(0. 1. 1)
3	4	(0. 1. 2)
3	6	(0. 1. 2)
5	2	(1. 0. 0)
5	4	(0. 0. 0)
5	6	(0. 0. 0)

CNN:

Historical and technical indicators		
Output length	Input Length	Optimal hyperparameters (Dropout. filters. kernel size. dense units)
1	2	(5. (32. 64. 128). 0.2. 128)

1	4	(5. (32. 64. 128). 0.4. 256)
1	6	(5. (32. 64. 128). 0.2. 128)
3	2	(3. (32. 64. 128). 0.4. 256)
3	4	(3. (32. 64. 128). 0.4. 256)
3	6	(3. (32. 64. 128). 0.4. 256)
5	2	(5. (32. 64. 128). 0.2. 128)
5	4	(3. (64. 128. 256). 0.3. 256)
5	6	(3. (32. 64. 128). 0.4. 64)

Historical indicators		
Output length	Input Length	Optimal hyperparameters (Dropout. filters. kernel size. dense units)
1	2	(3. (64. 128. 256). 0.3. 256)
1	4	(3. (32. 64. 128). 0.4. 64)
1	6	(5. (32. 64. 128). 0.2. 128)
3	2	(3. (32. 64. 128). 0.4. 64)
3	4	(5. (32. 64. 128). 0.4. 256)
3	6	(3. (32. 64. 128). 0.4. 64)
5	2	(3. (32. 64. 128). 0.4. 64)
5	4	(3. (64. 128. 256). 0.3. 256)
5	6	(3. (32. 64. 128). 0.4. 64)

Technical indicators		
Output length	Input Length	Optimal hyperparameters (Dropout. filters. kernel size. dense units)
1	2	(3. (32. 64. 128). 0.4. 256)
1	4	(5. (32. 64. 128). 0.2. 128)
1	6	(5. (32. 64. 128). 0.2. 128)
3	2	(3. (32. 64. 128). 0.4. 64)
3	4	(3. (32. 64. 128). 0.4. 64)
3	6	(5. (32. 64. 128). 0.2. 128)
5	2	(3. (32. 64. 128). 0.4. 64
5	4	(3. (64. 128. 256). 0.3. 256)
5	6	(3. (32. 64. 128). 0.4. 256)

Closing Price		
Output length	Input Length	Optimal hyperparameters (Dropout. filters. kernel size. dense units)
1	2	(5. (32. 64. 128). 0.2. 128)

1	4	(3. (32. 64. 128). 0.4. 256)
1	6	(5. (32. 64. 128). 0.2. 128)
3	2	(3. (32. 64. 128). 0.4. 256)
3	4	(3. (32. 64. 128). 0.4. 256)
3	6	(5. (32. 64. 128). 0.4. 256)
5	2	(3. (32. 64. 128). 0.4. 64)
5	4	(3. (64. 128. 256). 0.3. 256)
5	6	(5. (32. 64. 128). 0.4. 256)

Appendix D Confusion Matrices

Please note that all confusion matrices are averages of 3 seeds for LSTM and CNN.

LSTM

Historical and technical indicators				
Output length	Input Length	Confusion Matrix		
1	2	O/P	Uptrend	Downtrend
		Uptrend	57.33	28.67
		Downtrend	70.67	28.33
1	4	O/P	Uptrend	Downtrend
		Uptrend	0.67	48.67
		Downtrend	3.67	66.33
1	6	O/P	Uptrend	Downtrend
		Uptrend	0.00	46
		Downtrend	0.00	64.33
3	2	O/P	Uptrend	Downtrend
		Uptrend	24	39.67
		Downtrend	13.67	89.33
3	4	O/P	Uptrend	Downtrend
		Uptrend	0	80
		Downtrend	0	107
3	6	O/P	Uptrend	Downtrend
		Uptrend	0	80
		Downtrend	0	104
5	2	O/P	Uptrend	Downtrend
		Uptrend	21	43.33
		Downtrend	14.33	87.67
5	4	O/P	Uptrend	Downtrend
		Uptrend	0	82
		Downtrend	0	103
5	6	O/P	Uptrend	Downtrend
		Uptrend	0	81
		Downtrend	0	103

Technical indicators	

Output length	Input Length	Confusion Matrix		
1	2	O/P	Uptrend	Downtrend
		Uptrend	21	65
		Downtrend	22.30	76.70
1	4	O/P	Uptrend	Downtrend
		Uptrend	35.70	93
		Downtrend	29.30	122
1	6	O/P	Uptrend	Downtrend
		Uptrend	26.30	152
		Downtrend	29	156
3	2	O/P	Uptrend	Downtrend
		Uptrend	39	99.70
		Downtrend	22.70	90.30
3	4	O/P	Uptrend	Downtrend
		Uptrend	47	62.30
		Downtrend	30.70	76.70
3	6	O/P	Uptrend	Downtrend
		Uptrend	42.70	63.70
		Downtrend	25.70	78.30
5	2	O/P	Uptrend	Downtrend
		Uptrend	62.33	24.67
		Downtrend	13	85
5	4	O/P	Uptrend	Downtrend
		Uptrend	52	29.67
		Downtrend	16	87
5	6	O/P	Uptrend	Downtrend
		Uptrend	63	18
		Downtrend	13.67	89.33

Historical indicators				
Output length	Input Length	Validation Accuracy		
1	2	O/P	Uptrend	Downtrend
		Uptrend	0	72
		Downtrend	0	113
1	4	O/P	Uptrend	Downtrend
		Uptrend	0	87
		Downtrend	0	98
1	6	O/P	Uptrend	Downtrend
		Uptrend	0	81
		Downtrend	0	104
3	2	O/P	Uptrend	Downtrend
		Uptrend	69	0
		Downtrend	116	0
3	4	O/P	Uptrend	Downtrend
		Uptrend	0	80
		Downtrend	0	105
3	6	O/P	Uptrend	Downtrend
		Uptrend	0	73
		Downtrend	0	112

5	2	O/P	Uptrend	Downtrend
		Uptrend	14	61
		Downtrend	31	79
5	4	O/P	Uptrend	Downtrend
		Uptrend	5	73
		Downtrend	8	99
5	6	O/P	Uptrend	Downtrend
		Uptrend	0	82
		Downtrend	0	103

Closing Price				
Output length	Input Length	Confusion Matrix		
1	2	O/P	Uptrend	Downtrend
		Uptrend	0	72
		Downtrend	0	113
1	4	O/P	Uptrend	Downtrend
		Uptrend	13.50	73.60
		Downtrend	18.00	80.00
1	6	O/P	Uptrend	Downtrend
		Uptrend	0	81
		Downtrend	0	104
3	2	O/P	Uptrend	Downtrend
		Uptrend	0	69
		Downtrend	0	116
3	4	O/P	Uptrend	Downtrend
		Uptrend	29.00	51.00
		Downtrend	45.67	59.33
3	6	O/P	Uptrend	Downtrend
		Uptrend	0	73
		Downtrend	0	112
5	2	O/P	Uptrend	Downtrend
		Uptrend	0	72
		Downtrend	0	113
5	4	O/P	Uptrend	Downtrend
		Uptrend	9.00	78.00
		Downtrend	12.00	86.00
5	6	O/P	Uptrend	Downtrend
		Uptrend	0	81
		Downtrend	0	104

ARIMA:

Historical and technical indicators				
Output length	Input Length	Confusion Matrix		
1	2	O/P	Uptrend	Downtrend
		Uptrend	58	46
		Downtrend	40	41
1	4	O/P	Uptrend	Downtrend
		Uptrend	53	51
		Downtrend	43	38
1	6	O/P	Uptrend	Downtrend
		Uptrend	43	61
		Downtrend	33	47
3	2	O/P	Uptrend	Downtrend
		Uptrend	68	35
		Downtrend	25	57
3	4	O/P	Uptrend	Downtrend
		Uptrend	66	37
		Downtrend	30	52
3	6	O/P	Uptrend	Downtrend
		Uptrend	64	38
		Downtrend	29	53
5	2	O/P	Uptrend	Downtrend
		Uptrend	90	16
		Downtrend	24	55
5	4	O/P	Uptrend	Downtrend
		Uptrend	94	12
		Downtrend	28	51
5	6	O/P	Uptrend	Downtrend
		Uptrend	91	14
		Downtrend	29	50

Technical indicators				
Output length	Input Length	Confusion Matrix		
1	2	O/P	Uptrend	Downtrend
		Uptrend	79	25
		Downtrend	53	28
1	4	O/P	Uptrend	Downtrend
		Uptrend	77	27
		Downtrend	57	24
1	6	O/P	Uptrend	Downtrend
		Uptrend	64	40
		Downtrend	46	34
3	2	O/P	Uptrend	Downtrend
		Uptrend	78	25
		Downtrend	27	55

3	4	O/P	Uptrend	Downtrend
		Uptrend	75	28
		Downtrend	32	50
3	6	O/P	Uptrend	Downtrend
		Uptrend	74	28
		Downtrend	30	52
5	2	O/P	Uptrend	Downtrend
		Uptrend	86	20
		Downtrend	23	56
5	4	O/P	Uptrend	Downtrend
		Uptrend	88	18
		Downtrend	21	58
5	6	O/P	Uptrend	Downtrend
		Uptrend	88	17
		Downtrend	22	57

Historical indicators				
Output length	Input Length	Validation Accuracy		
1	2	O/P	Uptrend	Downtrend
		Uptrend	50	54
		Downtrend	42	39
1	4	O/P	Uptrend	Downtrend
		Uptrend	48	56
		Downtrend	41	40
1	6	O/P	Uptrend	Downtrend
		Uptrend	43	61
		Downtrend	29	52
3	2	O/P	Uptrend	Downtrend
		Uptrend	73	30
		Downtrend	29	53
3	4	O/P	Uptrend	Downtrend
		Uptrend	73	30
		Downtrend	26	56
3	6	O/P	Uptrend	Downtrend
		Uptrend	69	34
		Downtrend	29	53
5	2	O/P	Uptrend	Downtrend
		Uptrend	82	24
		Downtrend	23	56
5	4	O/P	Uptrend	Downtrend
		Uptrend	89	17
		Downtrend	26	53
5	6	O/P	Uptrend	Downtrend
		Uptrend	88	18
		Downtrend	21	58

Closing Price		
Output length	Input Length	Confusion Matrix

1	2	O/P	Uptrend	Downtrend
		Uptrend	12	92
		Downtrend	8	73
1	4	O/P	Uptrend	Downtrend
		Uptrend	11	93
		Downtrend	14	67
1	6	O/P	Uptrend	Downtrend
		Uptrend	12	92
		Downtrend	7	74
3	2	O/P	Uptrend	Downtrend
		Uptrend	64	39
		Downtrend	17	65
3	4	O/P	Uptrend	Downtrend
		Uptrend	18	85
		Downtrend	0	82
3	6	O/P	Uptrend	Downtrend
		Uptrend	38	65
		Downtrend	16	66
5	2	O/P	Uptrend	Downtrend
		Uptrend	60	46
		Downtrend	15	64
5	4	O/P	Uptrend	Downtrend
		Uptrend	80	26
		Downtrend	16	63
5	6	O/P	Uptrend	Downtrend
		Uptrend	82	24
		Downtrend	16	63

CNN:

Historical and technical indicators				
Output length	Input Length	Confusion Matrix		
1	2	O/P	Uptrend	Downtrend
		Uptrend	24	56
		Downtrend	30	75
1	4	O/P	Uptrend	Downtrend
		Uptrend	27	51
		Downtrend	38	67
1	6	O/P	Uptrend	Downtrend
		Uptrend	34	45
		Downtrend	47	55
3	2	O/P	Uptrend	Downtrend
		Uptrend	30	47
		Downtrend	42	64
3	4	O/P	Uptrend	Downtrend
		Uptrend	31	62
		Downtrend	34	61
3	6	O/P	Uptrend	Downtrend
		Uptrend	38	39
		Downtrend	34	69
5	2	O/P	Uptrend	Downtrend
		Uptrend	44.67	37.33
		Downtrend	27	76
5	4	O/P	Uptrend	Downtrend
		Uptrend	52.33	32.67
		Downtrend	18	81
5	6	O/P	Uptrend	Downtrend
		Uptrend	50.67	25.33
		Downtrend	26.67	80.33

Technical indicators				
Output length	Input Length	Confusion Matrix		
1	2	O/P	Uptrend	Downtrend
		Uptrend	28.67	58.33
		Downtrend	36	62
1	4	O/P	Uptrend	Downtrend
		Uptrend	6	84
		Downtrend	10.33	83.67
1	6	O/P	Uptrend	Downtrend
		Uptrend	26.33	64
		Downtrend	26	71
3	2	O/P	Uptrend	Downtrend
		Uptrend	22.67	58.33
		Downtrend	29.67	74

3	4	O/P	Uptrend	Downtrend
		Uptrend	17.67	70.67
		Downtrend	19.67	76.67
3	6	O/P	Uptrend	Downtrend
		Uptrend	28	53.67
		Downtrend	18.33	82
5	2	O/P	Uptrend	Downtrend
		Uptrend	48.33	29.67
		Downtrend	30.33	76.67
5	4	O/P	Uptrend	Downtrend
		Uptrend	48.33	29.67
		Downtrend	30.33	76.67
5	6	O/P	Uptrend	Downtrend
		Uptrend	23	54
		Downtrend	13	94

Historical indicators				
Output length	Input Length	Confusion Matrix		
1	2	O/P	Uptrend	Downtrend
		Uptrend	28.33	52
		Downtrend	37.67	67.33
1	4	O/P	Uptrend	Downtrend
		Uptrend	32.33	45.67
		Downtrend	51	55
1	6	O/P	Uptrend	Downtrend
		Uptrend	35.67	44.33
		Downtrend	50.67	52.33
3	2	O/P	Uptrend	Downtrend
		Uptrend	17.33	60.67
		Downtrend	31	75.33
3	4	O/P	Uptrend	Downtrend
		Uptrend	38.33	50.67
		Downtrend	44.33	50.67
3	6	O/P	Uptrend	Downtrend
		Uptrend	32.33	47.67
		Downtrend	33.67	69.33
5	2	O/P	Uptrend	Downtrend
		Uptrend	46.67	34.67
		Downtrend	25.67	78.67
5	4	O/P	Uptrend	Downtrend
		Uptrend	55.67	29.33
		Downtrend	24	75
5	6	O/P	Uptrend	Downtrend
		Uptrend	56	20
		Downtrend	30	77

Closing Price		
Output length	Input Length	Confusion Matrix

1	2	O/P	Uptrend	Downtrend
		Uptrend	0	87
		Downtrend	0	98
1	4	O/P	Uptrend	Downtrend
		Uptrend	0.6667	89.33
		Downtrend	0	94
1	6	O/P	Uptrend	Downtrend
		Uptrend	28	59
		Downtrend	23.33	73.76
3	2	O/P	Uptrend	Downtrend
		Uptrend	0	80
		Downtrend	0	105
3	4	O/P	Uptrend	Downtrend
		Uptrend	0	88
		Downtrend	0	96
3	6	O/P	Uptrend	Downtrend
		Uptrend	0	83
		Downtrend	0	101
5	2	O/P	Uptrend	Downtrend
		Uptrend	53	25
		Downtrend	26	81
5	4	O/P	Uptrend	Downtrend
		Uptrend	39	49.33
		Downtrend	11	84.67
5	6	O/P	Uptrend	Downtrend
		Uptrend	36.67	40
		Downtrend	21	85.33

Appendix E Results per indicator group and output/input length

LSTM:

Indicators	Output Length	Input Length	Mean Test Accuracy	F1-score	Standard Deviation
Historical and technical	1	2	0.4883	0.3815	0.0406
Historical and technical	1	4	0.5514	0.3669	0.0281
Historical and technical	1	6	0.5815	0.4046	0.0000
Historical and technical	3	2	0.5369	0.4234	0.1280
Historical and technical	3	4	0.5784	0.5676	0.0000
Historical and technical	3	6	0.5652	0.6054	0.0000
Historical and technical	5	2	0.5099	0.4524	0.0343
Historical and technical	5	4	0.5568	0.4349	0.0000

Historical and technical	5	6	0.5598	0.3982	0.0000
Technical	1	2	0.5279	0.4907	0.0352
Technical	1	4	0.5387	0.5096	0.0667
Technical	1	6	0.5707	0.5547	0.0109
Technical	3	2	0.6991	0.6941	0.0136
Technical	3	4	0.6541	0.6538	0.0236
Technical	3	6	0.6576	0.6526	0.0094
Technical	5	2	0.7964	0.8273	0.0083
Technical	5	4	0.7514	0.7476	0.0143
Technical	5	6	0.8279	0.7948	0.0220
Historical	1	2	0.5369	0.3815	0.1280
Historical	1	4	0.5297	0.3669	0.0000
Historical	1	6	0.5622	0.4046	0.0000
Historical	3	2	0.4876	0.4492	0.1014
Historical	3	4	0.5729	0.4110	0.0093
Historical	3	6	0.5959	0.4566	0.0164
Historical	5	2	0.5640	0.4534	0.0531
Historical	5	4	0.5730	0.4349	0.0094
Historical	5	6	0.5568	0.3982	0.0000
Closing price	1	2	0.5694	0.4678	0.0718
Closing price	1	4	0.5135	0.4005	0.0281
Closing price	1	6	0.5387	0.4339	0.0406
Closing price	3	2	0.6270	0.4833	0.0000
Closing price	3	4	0.4775	0.3567	0.0689
Closing price	3	6	0.6144	0.5169	0.0156
Closing price	5	2	0.7207	0.6072	0.0156
Closing price	5	4	0.4649	0.3823	0.1966
Closing price	5	6	0.3820	0.2896	0.0531

Average:

Indicators	Test accuracy	F1-score
Historical and technical	0.54757	0.4483
Technical	0.66930	0.6584
Historical	0.55322	0.3815
Closing price	0.54535	0.4376

ARIMA:

Indicators	Output Length	Input Length	Test Accuracy	F1-score	p-value
Historical and technical	1	2	0.5351	0.5872	0.1610
Historical and technical	1	4	0.4919	0.5153	0.0791
Historical and technical	1	6	0.4891	0.4483	0.3310
Historical and technical	3	2	0.6757	0.6603	0.1309
Historical and technical	3	4	0.6378	0.6364	0.0882
Historical and technical	3	6	0.6359	0.5882	0.0818
Historical and technical	5	2	0.7838	0.7353	0.1621
Historical and technical	5	4	0.7838	0.7398	0.1084
Historical and technical	5	6	0.7609	0.7238	0.0775
Technical	1	2	0.5784	0.4938	0.3513
Technical	1	4	0.5459	0.4513	0.4571
Technical	1	6	0.5326	0.4384	0.4358
Technical	3	2	0.7189	0.7342	0.3250
Technical	3	4	0.6757	0.6881	0.3732
Technical	3	6	0.6848	0.6939	0.4004
Technical	5	2	0.7676	0.7759	0.2939
Technical	5	4	0.7892	0.7778	0.3905
Technical	5	6	0.7880	0.7799	0.3896
Historical	1	2	0.4811	0.4286	0.2209
Historical	1	4	0.4757	0.4110	0.2026
Historical	1	6	0.5135	0.5556	0.1977
Historical	3	2	0.6811	0.6885	0.1384
Historical	3	4	0.6865	0.6842	0.1256
Historical	3	6	0.6432	0.6471	0.1523
Historical	5	2	0.7405	0.7848	0.1720
Historical	5	4	0.7351	0.7925	0.1965
Historical	5	6	0.7514	0.7778	0.1910
Closing price	1	2	0.4595	0.2330	0.2393
Closing price	1	4	0.4216	0.2254	0.3011
Closing price	1	6	0.4649	0.2330	0.3913
Closing price	3	2	0.6919	0.5613	0.0987
Closing price	3	4	0.5351	0.3846	0.1317
Closing price	3	6	0.5303	0.4640	0.2503
Closing price	5	2	0.6703	0.6472	0.2829
Closing price	5	4	0.7351	0.7805	0.0905
Closing price	5	6	0.7459	0.7746	0.1125

Average:

Indicators	Test accuracy	F1-score
Historical and technical	0.6438	0.6261
Technical	0.6757	0.6481
Historical	0.6342	0.6411
Closing price	0.5838	0.4782

CNN:

Indicators	Output Length	Input Length	Mean Test Accuracy	F1-score	Standard Deviation
Historical and technical	1	2	0.5171	0.6401	0.0255
Historical and technical	1	4	0.5217	0.5519	0.0193
Historical and technical	1	6	0.4754	0.5433	0.0134
Historical and technical	3	2	0.5117	0.5993	0.0287
Historical and technical	3	4	0.5236	0.5787	0.0102
Historical and technical	3	6	0.5829	0.5930	0.0297
Historical and technical	5	2	0.6739	0.7029	0.0067
Historical and technical	5	4	0.7192	0.7614	0.0112
Historical and technical	5	6	0.7158	0.7542	0.0248
Technical	1	2	0.5297	0.6011	0.0202
Technical	1	4	0.4565	0.5330	0.0203
Technical	1	6	0.4809	0.5224	0.0402
Technical	3	2	0.5171	0.6192	0.0270
Technical	3	4	0.4783	0.5183	0.0044
Technical	3	6	0.5191	0.6387	0.0178
Technical	5	2	0.6757	0.7272	0.0192
Technical	5	4	0.6975	0.7379	0.0068
Technical	5	6	0.6958	0.7549	0.0201
Historical	1	2	0.4919	0.5671	0.0117
Historical	1	4	0.4873	0.6395	0.0112
Historical	1	6	0.5362	0.6201	0.0112
Historical	3	2	0.5063	0.5998	0.0067
Historical	3	4	0.5362	0.6644	0.0026
Historical	3	6	0.5870	0.6721	0.0044
Historical	5	2	0.6667	0.7188	0.0092
Historical	5	4	0.6322	0.7340	0.0128
Historical	5	6	0.6431	0.7372	0.0051
Closing price	1	2	0.5297	0.6926	0.0000
Closing price	1	4	0.5163	0.6776	0.0044
Closing price	1	6	0.5326	0.6407	0.0160
Closing price	3	2	0.5676	0.7241	0.0000
Closing price	3	4	0.5217	0.6857	0.0000
Closing price	3	6	0.5399	0.7088	0.0128
Closing price	5	2	0.7369	0.7810	0.0067

Closing price	5	4	0.6467	0.7486	0.0177
Closing price	5	6	0.6359	0.7308	0.0117

Average:

Indicators	Test accuracy	F1-score
Historical and technical	0.58237	0.6361
Technical	0.56118	0.6281
Historical	0.56522	0.6614
Closing price	0.58082	0.7080

Appendix F Simulation trades

Day	Trade Type	Amount Invested	Closing Price	Days Until Cash Out	Balance Change	Ending Balance
2	Buy	200	13897.29981	5	208.1824051	1008.182
2	Sell	200	13897.29981	0	200	1200
2	Buy	200	13897.29981	5	208.1824051	1008.182
3	Buy	200	13711	5	205.6800092	1013.862
3	Sell	200	13711	0	200	1400
3	Buy	200	13711	5	205.6800092	1013.862

4	Buy	200	13411.95996	5	196.9499945	1010.812
4	Sell	200	13411.95996	0	200	1600
4	Sell	200	13411.95996	0	200	1013.862
5	Buy	200	13371.57031	5	198.7883807	1009.601
5	Buy	200	13371.57031	5	198.7883807	1598.788
5	Sell	200	13371.57031	0	200	1013.862
6	Buy	200	13643.58984	5	207.1188144	1016.72
6	Buy	200	13643.58984	5	207.1188144	1605.907
6	Sell	200	13643.58984	0	200	1013.862
7	Buy	200	13351.08008	5	207.9722483	1024.692
7	Sell	200	13351.08008	0	200	1805.907
7	Buy	200	13351.08008	5	207.9722483	1021.835
8	Buy	200	13332.36035	5	205.0367479	1029.729
8	Sell	200	13332.36035	0	200	2005.907
8	Sell	200	13332.36035	0	200	1021.835
9	Buy	200	13619.66016	5	218.0761092	1047.805
9	Sell	200	13619.66016	0	200	2205.907
9	Sell	200	13619.66016	0	200	1021.835
10	Buy	200	13453.07031	5	215.439924	1063.245
10	Sell	200	13453.07031	0	200	2405.907
10	Buy	200	13453.07031	5	215.439924	1037.275
11	Buy	200	13174.65039	5	204.7099313	1067.955
11	Sell	200	13174.65039	0	200	2605.907
11	Sell	200	13174.65039	0	200	1037.275
12	Buy	200	12839.29004	5	208.1826532	1076.137
12	Sell	200	12839.29004	0	200	2805.907
12	Sell	200	12839.29004	0	200	1037.275

13	Buy	200	13004.84961	5	207.4797279	1083.617
13	Sell	200	13004.84961	0	200	3005.907
13	Sell	200	13004.84961	0	200	1037.275
14	Buy	200	12490.74023	5	198.8376166	1082.455
14	Sell	200	12490.74023	0	200	3205.907
14	Buy	200	12490.74023	5	198.8376166	1036.112
15	Buy	200	12488.92969	5	192.6581444	1075.113
15	Buy	200	12488.92969	5	192.6581444	3198.565
15	Sell	200	12488.92969	0	200	1036.112
16	Buy	200	12871.53027	5	208.9925923	1084.105
16	Buy	200	12871.53027	5	208.9925923	3207.558
16	Sell	200	12871.53027	0	200	1036.112
17	Buy	200	12334.63965	5	203.1286094	1087.234
17	Sell	200	12334.63965	0	200	3407.558
17	Buy	200	12334.63965	5	203.1286094	1039.241
18	Buy	200	12536.01953	5	215.7059262	1102.94
18	Sell	200	12536.01953	0	200	3607.558
18	Sell	200	12536.01953	0	200	1039.241
19	Buy	200	12563.75977	5	214.0758745	1117.016
19	Sell	200	12563.75977	0	200	3807.558
19	Buy	200	12563.75977	5	214.0758745	1053.317
20	Sell	200	12964.86035	0	200	1317.016
20	Sell	200	12964.86035	0	200	4007.558
20	Buy	200	12964.86035	5	228.1694171	1081.486
21	Buy	200	12317.69043	5	216.6517246	1333.667
21	Sell	200	12317.69043	0	200	4207.558
21	Buy	200	12317.69043	5	216.6517246	1098.138

22	Sell	200	12144.66016	0	200	1533.667
22	Sell	200	12144.66016	0	200	4407.558
22	Sell	200	12144.66016	0	200	1098.138
23	Buy	200	11623.25	5	199.3219455	1532.989
23	Sell	200	11623.25	0	200	4607.558
23	Sell	200	11623.25	0	200	1098.138
24	Buy	200	11737.66992	5	195.8805256	1528.87
24	Buy	200	11737.66992	5	195.8805256	4603.438
24	Sell	200	11737.66992	0	200	1098.138
25	Buy	200	11364.24023	5	199.0557112	1527.926
25	Buy	200	11364.24023	5	199.0557112	4602.494
25	Buy	200	11364.24023	5	199.0557112	1097.194
26	Buy	200	11370.95996	5	199.6919693	1527.618
26	Buy	200	11370.95996	5	199.6919693	4602.186
26	Sell	200	11370.95996	0	200	1097.194
27	Buy	200	11805	5	207.9329802	1535.551
27	Buy	200	11805	5	207.9329802	4610.119
27	Buy	200	11805	5	207.9329802	1105.127
28	Sell	200	11662.79004	0	200	1735.551
28	Sell	200	11662.79004	0	200	4810.119
28	Buy	200	11662.79004	5	202.2109671	1107.337
29	Buy	200	11984.51953	5	212.784811	1748.335
29	Sell	200	11984.51953	0	200	5010.119
29	Sell	200	11984.51953	0	200	1107.337
30	Sell	200	11418.15039	0	200	1948.335
30	Sell	200	11418.15039	0	200	5210.119
30	Buy	200	11418.15039	5	199.7098344	1107.047

31	Buy	200	11388.5	5	194.0011775	1942.337
31	Buy	200	11388.5	5	194.0011775	5204.12
31	Sell	200	11388.5	0	200	1107.047
32	Buy	200	11354.62012	5	187.1980636	1929.535
32	Buy	200	11354.62012	5	187.1980636	5191.318
32	Sell	200	11354.62012	0	200	1107.047
33	Buy	200	11535.26953	5	190.9593162	1920.494
33	Buy	200	11535.26953	5	190.9593162	5182.278
33	Sell	200	11535.26953	0	200	1107.047
34	Sell	200	11264.4502	0	200	2120.494
34	Buy	200	11264.4502	5	187.8275509	5170.105
34	Buy	200	11264.4502	5	187.8275509	1094.875
35	Buy	200	11434.74023	5	185.6756143	2106.17
35	Buy	200	11434.74023	5	185.6756143	5155.781
35	Sell	200	11434.74023	0	200	1094.875
36	Sell	200	11740.65039	0	200	2306.17
36	Buy	200	11740.65039	5	195.4701376	5151.251
36	Buy	200	11740.65039	5	195.4701376	1090.345
37	Buy	200	12131.12988	5	201.156747	2307.326
37	Buy	200	12131.12988	5	201.156747	5152.408
37	Buy	200	12131.12988	5	201.156747	1091.502
38	Sell	200	12081.38965	0	200	2507.326
38	Sell	200	12081.38965	0	200	5352.408
38	Buy	200	12081.38965	5	198.4584962	1089.96
39	Buy	200	11994.45996	5	198.4807625	2505.807
39	Buy	200	11994.45996	5	198.4807625	5350.888
39	Sell	200	11994.45996	0	200	1089.96

40	Sell	200	12316.90039	0	200	2705.807
40	Buy	200	12316.90039	5	209.5739134	5360.462
40	Sell	200	12316.90039	0	200	1089.96
41	Buy	200	12012.73047	5	211.86437	2717.671
41	Sell	200	12012.73047	0	200	5560.462
41	Buy	200	12012.73047	5	211.86437	1101.825
42	Sell	200	12061.37012	0	200	2917.671
42	Sell	200	12061.37012	0	200	5760.462
42	Sell	200	12061.37012	0	200	1101.825
43	Buy	200	12175.23047	5	224.8769371	2942.548
43	Sell	200	12175.23047	0	200	5960.462
43	Buy	200	12175.23047	5	224.8769371	1126.702
44	Sell	200	12086.26953	0	200	3142.548
44	Sell	200	12086.26953	0	200	6160.462
44	Buy	200	12086.26953	5	217.7872919	1144.489
45	Buy	200	11754.23047	5	220.8175933	3163.366
45	Sell	200	11754.23047	0	200	6360.462
45	Sell	200	11754.23047	0	200	1144.489
46	Buy	200	11340.01953	5	210.0324576	3173.398
46	Sell	200	11340.01953	0	200	6560.462
46	Sell	200	11340.01953	0	200	1144.489
47	Buy	200	10809.23047	5	195.3010698	3168.699
47	Sell	200	10809.23047	0	200	6760.462
47	Sell	200	10809.23047	0	200	1144.489
48	Buy	200	10828.34961	5	195.9336137	3164.633
48	Buy	200	10828.34961	5	195.9336137	6756.396
48	Sell	200	10828.34961	0	200	1144.489

49	Buy	200	11099.15039	5	197.6310936	3162.264
49	Buy	200	11099.15039	5	197.6310936	6754.027
49	Buy	200	11099.15039	5	197.6310936	1142.12
50	Buy	200	10646.09961	5	183.4329432	3145.697
50	Buy	200	10646.09961	5	183.4329432	6737.46
50	Buy	200	10646.09961	5	183.4329432	1125.553
51	Buy	200	10798.34961	5	187.3973351	3133.094
51	Buy	200	10798.34961	5	187.3973351	6724.857
51	Sell	200	10798.34961	0	200	1125.553
52	Buy	200	11069.29981	5	197.9924012	3131.087
52	Buy	200	11069.29981	5	197.9924012	6722.85
52	Buy	200	11069.29981	5	197.9924012	1123.545
53	Buy	200	11053.08008	5	197.7668491	3128.854
53	Buy	200	11053.08008	5	197.7668491	6720.617
53	Buy	200	11053.08008	5	197.7668491	1121.312
54	Buy	200	11232.19043	5	203.6894549	3132.543
54	Buy	200	11232.19043	5	203.6894549	6724.306
54	Sell	200	11232.19043	0	200	1121.312
55	Buy	200	11607.62012	5	208.6228791	3141.166
55	Sell	200	11607.62012	0	200	6924.306
55	Buy	200	11607.62012	5	208.6228791	1129.935
56	Sell	200	11524.54981	0	200	3341.166
56	Sell	200	11524.54981	0	200	7124.306
56	Buy	200	11524.54981	5	203.5736668	1133.509
57	Buy	200	11181.54004	5	196.8260525	3337.992
57	Sell	200	11181.54004	0	200	7324.306
57	Sell	200	11181.54004	0	200	1133.509

58	Buy	200	11177.88965	5	192.3681848	3330.36
58	Buy	200	11177.88965	5	192.3681848	7316.674
58	Sell	200	11177.88965	0	200	1133.509
59	Buy	200	11028.74023	5	189.5736451	3319.934
59	Buy	200	11028.74023	5	189.5736451	7306.248
59	Sell	200	11028.74023	0	200	1133.509
60	Buy	200	11127.84961	5	195.695795	3315.63
60	Buy	200	11127.84961	5	195.695795	7301.944
60	Sell	200	11127.84961	0	200	1133.509
61	Buy	200	11322.24023	5	201.0210589	3316.651
61	Buy	200	11322.24023	5	201.0210589	7302.965
61	Buy	200	11322.24023	5	201.0210589	1134.53
62	Buy	200	11361.84961	5	202.0318954	3318.683
62	Sell	200	11361.84961	0	200	7502.965
62	Buy	200	11361.84961	5	202.0318954	1136.562
63	Buy	200	11621.34961	5	206.5799114	3325.263
63	Sell	200	11621.34961	0	200	7702.965
63	Buy	200	11621.34961	5	206.5799114	1143.142
64	Buy	200	11635.30957	5	203.193904	3328.457
64	Sell	200	11635.30957	0	200	7902.965
64	Buy	200	11635.30957	5	203.193904	1146.335
65	Buy	200	11372.59961	5	200.2209463	3328.677
65	Sell	200	11372.59961	0	200	8102.965
65	Buy	200	11372.59961	5	200.2209463	1146.556
66	Sell	200	11264.73047	0	200	3528.677
66	Sell	200	11264.73047	0	200	8302.965
66	Sell	200	11264.73047	0	200	1146.556

67	Buy	200	11247.58008	5	189.0722909	3517.75
67	Buy	200	11247.58008	5	189.0722909	8292.037
67	Sell	200	11247.58008	0	200	1146.556
68	Buy	200	11251.19043	5	186.5929346	3504.343
68	Buy	200	11251.19043	5	186.5929346	8278.63
68	Sell	200	11251.19043	0	200	1146.556
69	Buy	200	11452.41992	5	193.5493177	3497.892
69	Buy	200	11452.41992	5	193.5493177	8272.179
69	Buy	200	11452.41992	5	193.5493177	1140.106
70	Buy	200	11360.04981	5	192.8264117	3490.718
70	Buy	200	11360.04981	5	192.8264117	8265.006
70	Buy	200	11360.04981	5	192.8264117	1132.932
71	Buy	200	11713.15039	5	202.6046125	3493.323
71	Buy	200	11713.15039	5	202.6046125	8267.61
71	Sell	200	11713.15039	0	200	1132.932
72	Buy	200	11897.65039	5	197.7598932	3491.083
72	Sell	200	11897.65039	0	200	8467.61
72	Buy	200	11897.65039	5	197.7598932	1130.692
73	Buy	200	12059.61035	5	198.306619	3489.39
73	Buy	200	12059.61035	5	198.306619	8465.917
73	Buy	200	12059.61035	5	198.306619	1128.999
74	Buy	200	11834.11035	5	191.0161572	3480.406
74	Buy	200	11834.11035	5	191.0161572	8456.933
74	Buy	200	11834.11035	5	191.0161572	1120.015
75	Buy	200	11782.66992	5	190.5196625	3470.925
75	Buy	200	11782.66992	5	190.5196625	8447.453
75	Sell	200	11782.66992	0	200	1120.015

76	Sell	200	11562.57031	0	200	3670.925
76	Buy	200	11562.57031	5	187.2669083	8434.72
76	Sell	200	11562.57031	0	200	1120.015
77	Buy	200	12032.41992	5	189.9631797	3660.889
77	Buy	200	12032.41992	5	189.9631797	8424.683
77	Sell	200	12032.41992	0	200	1120.015
78	Buy	200	12162.58984	5	191.2269687	3652.115
78	Buy	200	12162.58984	5	191.2269687	8415.91
78	Buy	200	12162.58984	5	191.2269687	1111.242
79	Buy	200	12390.69043	5	195.783396	3647.899
79	Buy	200	12390.69043	5	195.783396	8411.693
79	Buy	200	12390.69043	5	195.783396	1107.025
80	Buy	200	12368.98047	5	195.6426847	3643.542
80	Buy	200	12368.98047	5	195.6426847	8407.336
80	Buy	200	12368.98047	5	195.6426847	1102.668
81	Buy	200	12348.75977	5	197.6761527	3641.218
81	Buy	200	12348.75977	5	197.6761527	8405.012
81	Sell	200	12348.75977	0	200	1102.668
82	Sell	200	12668.16016	0	200	3841.218
82	Buy	200	12668.16016	5	197.0961874	8402.108
82	Sell	200	12668.16016	0	200	1102.668
83	Buy	200	12720.58008	5	199.0715102	3840.289
83	Buy	200	12720.58008	5	199.0715102	8401.18
83	Buy	200	12720.58008	5	199.0715102	1101.739
84	Sell	200	12657.54981	0	200	4040.289
84	Buy	200	12657.54981	5	194.0272103	8395.207
84	Buy	200	12657.54981	5	194.0272103	1095.767

85	Buy	200	12644.45996	5	192.6327238	4032.922
85	Buy	200	12644.45996	5	192.6327238	8387.84
85	Sell	200	12644.45996	0	200	1095.767
86	Sell	200	12493.92969	0	200	4232.922
86	Buy	200	12493.92969	5	190.7098981	8378.55
86	Sell	200	12493.92969	0	200	1095.767
87	Buy	200	12854.79981	5	198.7120183	4231.634
87	Buy	200	12854.79981	5	198.7120183	8377.262
87	Sell	200	12854.79981	0	200	1095.767
88	Sell	200	12779.91016	0	200	4431.634
88	Buy	200	12779.91016	5	197.1396093	8374.401
88	Buy	200	12779.91016	5	197.1396093	1092.906
89	Buy	200	13047.19043	5	205.3831529	4437.017
89	Buy	200	13047.19043	5	205.3831529	8379.784
89	Sell	200	13047.19043	0	200	1092.906
90	Sell	200	13128.04981	0	200	4637.017
90	Sell	200	13128.04981	0	200	8579.784
90	Buy	200	13128.04981	5	212.0579131	1104.964
91	Buy	200	13102.54981	5	211.6506346	4648.668
91	Sell	200	13102.54981	0	200	8779.784
91	Buy	200	13102.54981	5	211.6506346	1116.615
92	Sell	200	12938.12012	0	200	4848.668
92	Sell	200	12938.12012	0	200	8979.784
92	Sell	200	12938.12012	0	200	1116.615
93	Buy	200	12965.33984	5	205.1596386	4853.827
93	Sell	200	12965.33984	0	200	9179.784
93	Sell	200	12965.33984	0	200	1116.615

94	Buy	200	12705.21973	5	209.2822142	4863.11
94	Sell	200	12705.21973	0	200	9379.784
94	Buy	200	12705.21973	5	209.2822142	1125.897
95	Buy	200	12381.57031	5	206.056089	4869.166
95	Sell	200	12381.57031	0	200	9579.784
95	Sell	200	12381.57031	0	200	1125.897
96	Buy	200	12381.29981	5	208.3843188	4877.55
96	Sell	200	12381.29981	0	200	9779.784
96	Sell	200	12381.29981	0	200	1125.897
97	Buy	200	12431.53027	5	210.4150246	4887.965
97	Sell	200	12431.53027	0	200	9979.784
97	Sell	200	12431.53027	0	200	1125.897
98	Buy	200	12639.26953	5	214.4952098	4902.46
98	Sell	200	12639.26953	0	200	10179.78
98	Buy	200	12639.26953	5	214.4952098	1140.392
99	Buy	200	12141.70996	5	208.7843821	4911.245
99	Sell	200	12141.70996	0	200	10379.78
99	Buy	200	12141.70996	5	208.7843821	1149.177
100	Buy	200	12017.66992	5	208.1899254	4919.435
100	Sell	200	12017.66992	0	200	10579.78
100	Sell	200	12017.66992	0	200	1149.177
101	Buy	200	11883.13965	5	201.5474903	4920.982
101	Sell	200	11883.13965	0	200	10779.78
101	Sell	200	11883.13965	0	200	1149.177
102	Buy	200	11816.2002	5	199.2256081	4920.208
102	Sell	200	11816.2002	0	200	10979.78
102	Sell	200	11816.2002	0	200	1149.177

103	Buy	200	11785.12988	5	194.5975673	4914.805
103	Buy	200	11785.12988	5	194.5975673	10974.38
103	Sell	200	11785.12988	0	200	1149.177
104	Buy	200	11630.86035	5	189.6375582	4904.443
104	Buy	200	11630.86035	5	189.6375582	10964.02
104	Sell	200	11630.86035	0	200	1149.177
105	Buy	200	11544.91016	5	198.4757877	4902.919
105	Buy	200	11544.91016	5	198.4757877	10962.5
105	Sell	200	11544.91016	0	200	1149.177
106	Buy	200	11791.90039	5	201.2324689	4904.151
106	Buy	200	11791.90039	5	201.2324689	10963.73
106	Sell	200	11791.90039	0	200	1149.177
107	Buy	200	11862.12988	5	205.3628786	4909.514
107	Sell	200	11862.12988	0	200	11163.73
107	Buy	200	11862.12988	5	205.3628786	1154.539
108	Sell	200	12112.30957	0	200	5109.514
108	Sell	200	12112.30957	0	200	11363.73
108	Buy	200	12112.30957	5	211.5982872	1166.138
109	Buy	200	12266.41016	5	212.6812204	5122.195
109	Sell	200	12266.41016	0	200	11563.73
109	Buy	200	12266.41016	5	212.6812204	1178.819
110	Sell	200	11633.57031	0	200	5322.195
110	Sell	200	11633.57031	0	200	11763.73
110	Buy	200	11633.57031	5	203.6502337	1182.469
111	Buy	200	11719.67969	5	208.9034007	5331.099
111	Sell	200	11719.67969	0	200	11963.73
111	Sell	200	11719.67969	0	200	1182.469

112	Sell	200	11552.36035	0	200	5531.099
112	Sell	200	11552.36035	0	200	12163.73
112	Buy	200	11552.36035	5	208.7749008	1191.244
113	Buy	200	11448.40039	5	210.6822683	5541.781
113	Sell	200	11448.40039	0	200	12363.73
113	Sell	200	11448.40039	0	200	1191.244
114	Buy	200	11535.01953	5	213.5537357	5555.335
114	Sell	200	11535.01953	0	200	12563.73
114	Sell	200	11535.01953	0	200	1191.244
115	Buy	200	11425.04981	5	210.9986575	5566.333
115	Sell	200	11425.04981	0	200	12763.73
115	Buy	200	11425.04981	5	210.9986575	1202.243
116	Buy	200	11220.19043	5	203.0502403	5569.383
116	Sell	200	11220.19043	0	200	12963.73
116	Sell	200	11220.19043	0	200	1202.243
117	Buy	200	11066.80957	5	206.1336345	5575.517
117	Sell	200	11066.80957	0	200	13163.73
117	Sell	200	11066.80957	0	200	1202.243
118	Buy	200	10867.92969	5	205.5279892	5581.045
118	Sell	200	10867.92969	0	200	13363.73
118	Sell	200	10867.92969	0	200	1202.243
119	Buy	200	10802.91992	5	199.7686682	5580.814
119	Sell	200	10802.91992	0	200	13563.73
119	Sell	200	10802.91992	0	200	1202.243
120	Buy	200	10829.5	5	193.7921005	5574.606
120	Buy	200	10829.5	5	193.7921005	13557.52
120	Sell	200	10829.5	0	200	1202.243

121	Buy	200	11051.63965	5	198.2598774	5572.866
121	Buy	200	11051.63965	5	198.2598774	13555.78
121	Buy	200	11051.63965	5	198.2598774	1200.503